

SECTION 9 - SHIMMING



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.
- HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.

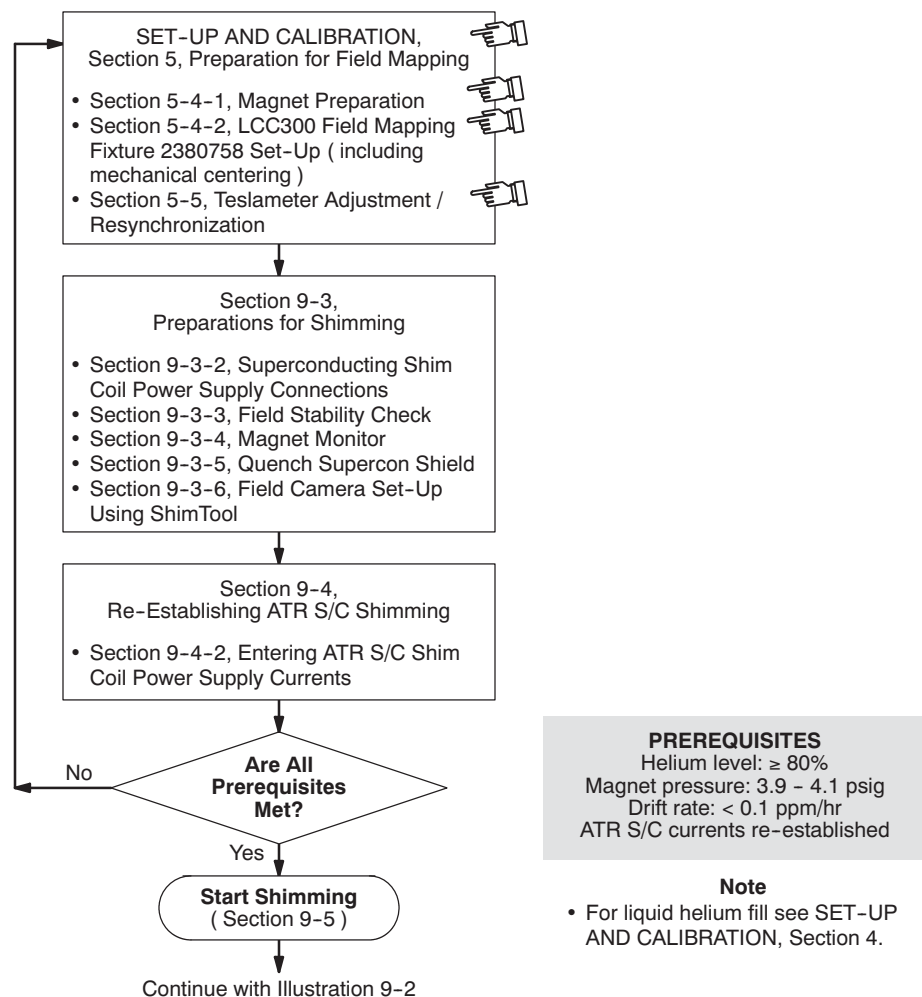


DO NOT PERFORM THIS PROCEDURE WITHOUT ASSISTANCE OF A QUALIFIED “WORK ASSISTANT.”

9-1 INTRODUCTION

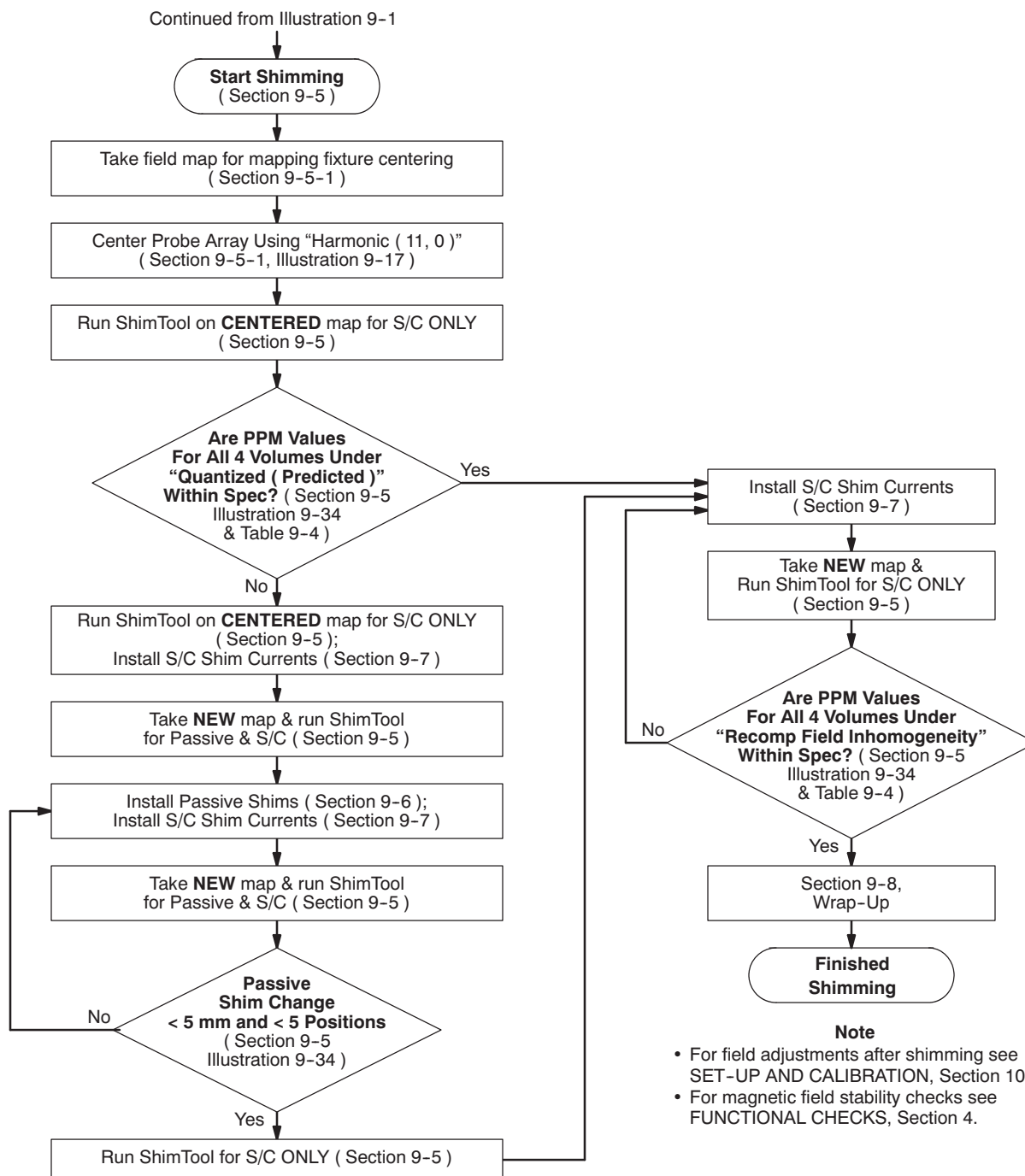
“Shimming” is the process of measuring the magnet’s field over a fixed volume and then calculating and installing active (superconducting) and passive (steel) shims to improve field homogeneity. LCC300 magnets factory shimming results are in the Acceptance Test Report (ATR) shipped with the magnet. On-site shimming is performed to compensate for unequal distribution of room steel that’s effecting field homogeneity. Additional shimming may be required following any significant environmental changes near the magnet. It is important to perform shimming after the magnet has been sited and ramped and the field stabilized.

After factory superconducting (S/C) shim currents are reapplied to the magnet, use ShimTool software to calculate any additional passive and S/C (active) shimming required on site. Flowcharts of shimming prerequisites and the shimming process are shown in Illustrations 9-1 and 9-2, respectively.



SHIMMING PROCESS FLOWCHART, BEFORE SHIMMING
 ILLUSTRATION 9-1

9-1 INTRODUCTION (continued)



SHIMMING PROCESS FLOWCHART, SHIMMING
ILLUSTRATION 9-2

9-2 TOOLS / EQUIPMENT

WARNING!

ALWAYS INSPECT ALL COMPONENTS OF ALL TOOLS / EQUIPMENT BEFORE USAGE. NEVER USE TOOLS / EQUIPMENT THAT CANNOT BE USED SAFELY.

9-2-1 Safety Equipment

- Portable Oxygen Monitor 2287000
- 3.0T Warning Sign & Label Kit 2379494
- Cryogen Safety Wear Kit 46-271137G1
- Nonabsorbent protective clothing (long sleeve shirt and pants)
- Nonferrous Shoes

9-2-2 Other Tools / Equipment

- LCC300 Shim Camera Kit 2386028
- For Passive Shimming:
 - LCC300 Passive Shimming Kit 2386029
(includes LCC300 Field Mapping Fixture 2380758)
 - LCC300 Passive Shimming Pull Assist Kit 5104145
 - 3/8 Inch Drive Electric Drill
- For re-establishing ATR S/C shimming:
 - MetroLab Teslameter Kit 46-251865G4
 - #6 Teslameter Probe 2295525-5
 - #5 MetroLab Probe 46-255839P104
- Nonmagnetic Tools Kit 46-320273G4
- Stopwatch or other timer
- 5 - 10 Gallon (20 - 40 Liter) Bucket
- Source of compressed gas (air or nitrogen preferred)

Note

The bucket and compressed gas are used to blow out the TRM coolant lines. If those lines must be drained at the magnet end, then the bucket and compressed gas hose / container **MUST** be nonmagnetic.

9-2 TOOLS / EQUIPMENT (continued)

9-2-3 Software

- GE ShimTool Software, Version 2.5a (5102783; included in LCC300 Shim Camera Kit 2386028)

9-3 PREPARATIONS FOR SHIMMING

9-3-1 Introduction

The preparations covered here are required before shimming the magnet.

1. Perform SET-UP AND CALIBRATION, Section 5-4, Preparation for Field Mapping, and Section 5-5, Teslameter Adjustment / Resynchronization, before shimming, if not already done.
2. Allow the magnet to stabilize to < 0.1 ppm/hr (2 Hz/hr measured using a single number 6 probe) main field drift before shimming. This will take 12 hours. Other commissioning functions can be performed during this time. The minimum helium level to start shimming is 80%.

IMPORTANT !!!

Make sure the "C0BO" (Controlled Zero Boil-Off) Heater on the Magnet Monitor is turned on.



Moving articles or equipment in the magnet room may affect field readings.

IMPORTANT !!!

Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.

9-3-2 Superconducting Shim Coil Power Supply Connections

IMPORTANT !!!

Only use nonmagnetic tools when working in the magnet's field.



Do not rest Shim Lead Pigtail Cable (2135362) on Coldhead or Coldhead Sleeve.

1. Verify that the Shim Lead Pigtail Cable (2135362) is connected to P1 -A on the Shim Lead Assembly. See Illustration 9-3.

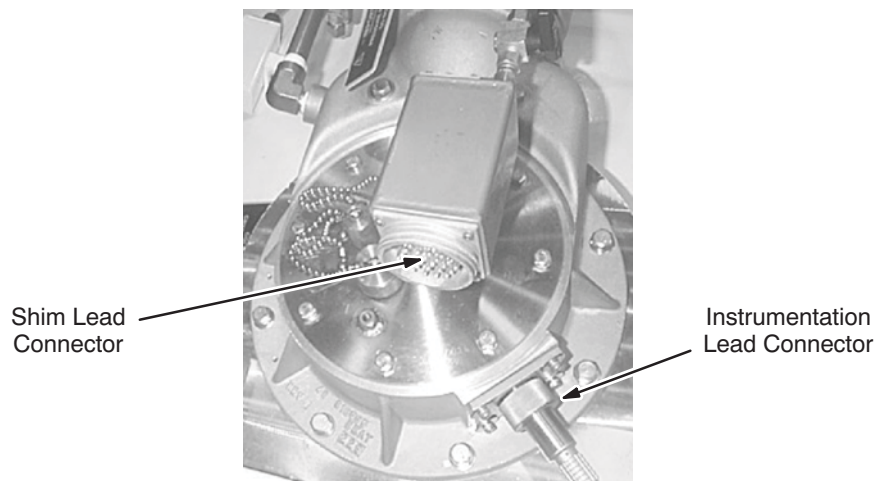
9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-2 Superconducting Shim Coil Power Supply Connections (continued)****MAGNET ELECTRICAL CONNECTIONS**

ILLUSTRATION 9-3

2. Make sure the Shim Lead Assembly is engaged in conformance to SET-UP AND CALIBRATION, Section 6-3. The valve on the Shim Lead Exhaust Fitting must be opened to increase current carrying ability in the Shim Leads.

WARNING!

DO NOT CONNECT INPUT POWER TO OR TURN ON THE SUPERCONDUCTING SHIM COIL POWER SUPPLY UNTIL ALL CURRENT CONTROLS ARE VERIFIED TO BE SET AT ZERO (FULLY COUNTERCLOCKWISE). TURNING ON THE SUPERCONDUCTING SHIM COIL POWER SUPPLY WHEN ANY CURRENT CONTROL IS NOT AT ZERO MAY CAUSE IRREPARABLE DAMAGE TO THE VAPOR-COOLED SHIM COIL LEADS.

CAUTION

Equipment damage is possible if heater currents are not set correctly.

3. Connect power to the Superconducting Shim Coil Power Supply and make sure the heater currents are $T1 = 610 \pm 5$ mA, $T2 = 610 \pm 5$ mA and Axial = 710 ± 10 mA. See Section 4 of the vendor's service manual for details.

Note

Always check ATR for any possible magnet-specific current settings.

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-2 Superconducting Shim Coil Power Supply Connections (continued)**

4. After sufficient cooling of the Shim Lead Assembly, verify that all six shim power supply channels operate at both positive and negative polarities for Axial, Transverse 1 and Transverse 2 shim groups. See Illustration 9-4.

Note

This also verifies the connection continuity through the Shim Leads.

5. Disconnect input power to the Superconducting Shim Coil Power Supply.

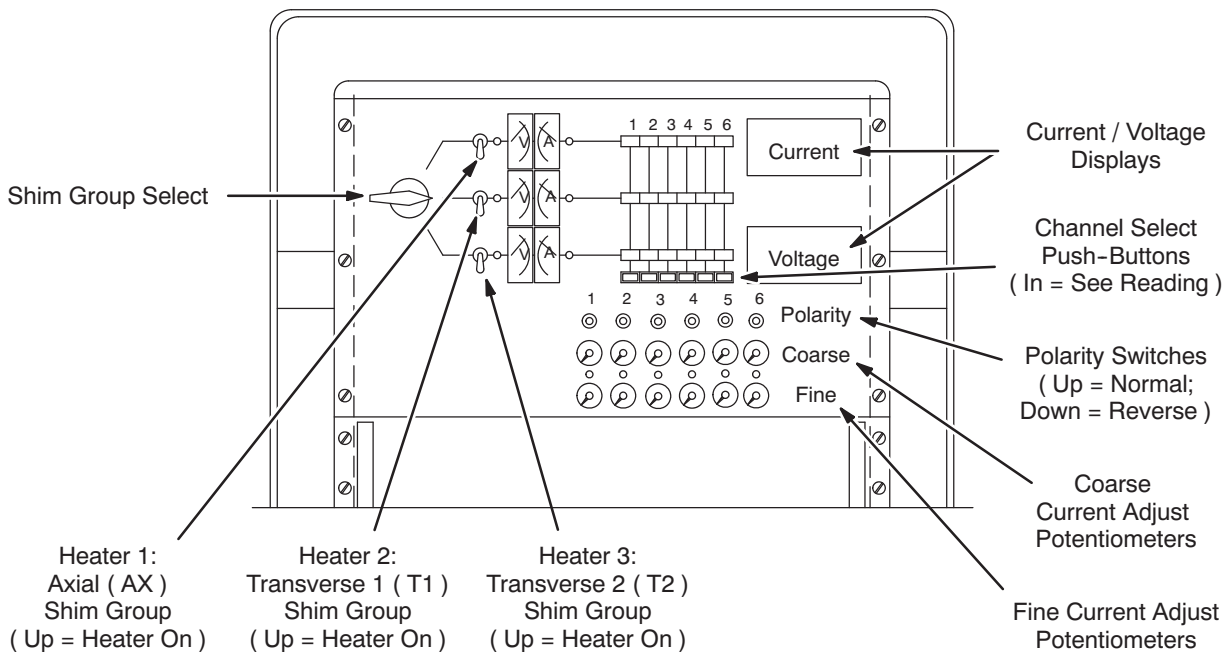
**SHIM POWER SUPPLY CONTROLS (FRONT PANEL)**

ILLUSTRATION 9-4

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-2 Superconducting Shim Coil Power Supply Connections (continued)**

6. Verify that P2 and P2W on the Superconducting Shim Coil Cable (RUN 603; 46-260726G1) are connected to the J2 and J2-W Shim Output Connectors on the back of the S/C Shim Coil Power Supply (MS7-A1). See Illustration 9-5.

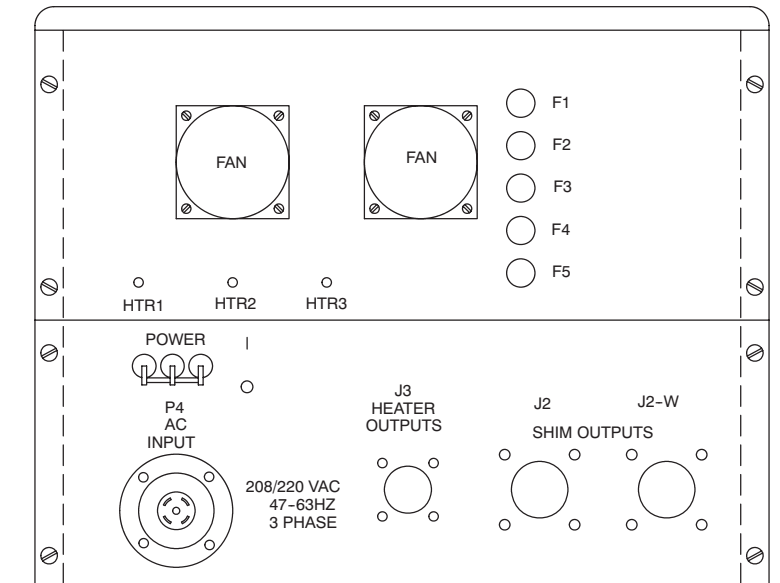
**SHIM POWER SUPPLY OUTPUT CONNECTIONS (BACK PANEL)**

ILLUSTRATION 9-5

7. Verify that J2 and J2-W on the other end of the Superconducting Shim Coil Cable are connected to the Shim Lead Pigtail Cable (2135362).
8. Connect P703 on the Switch Heater Cable (RUN 604; 46-260724G1) to connector J3 on the back of the S/C Shim Coil Power Supply (MS7-A1).



Only one Switch Heater Cable is to be connected to the Shim Lead Assembly.

9. Verify that connector P5 on the Switch Heater Cable is connected to J5-1 or J5-2 on the Shim Lead Assembly.
10. Verify using DATA SHEETS, Section 8, S/C Shimming Data, and any SHM file printouts (as in Illustration 9-33) available that no T1 or T2 currents have been loaded. Then turn on the T1, T2 and Axial heaters as required to increase magnet pressure to 4.0 ± 0.1 psig. Turn the heaters off when pressure is reached.

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-3 Field Stability Check****Note**

This check requires 12 hours between the first measurement and the second.

WARNING!

BEFORE TURNING ON THE SUPERCONDUCTING SHIM COIL POWER SUPPLY, MAKE SURE ALL CURRENT CONTROLS ARE SET TO ZERO (FULL COUNTERCLOCKWISE).

1. Connect input power to the S/C Shim Coil Power Supply and turn on ("I") all Shim Coil Switch Heaters for five minutes to dump any residual Shim Currents.
2. Turn off ("O") the Switch Heaters. Allow 3 to 5 minutes for the switch heaters to go persistent before continuing with Step 3.
3. Disconnect the heater cable (P703) from the Shim Power Supply.
4. Turn off the Shim Power Supply and disconnect the power input cable.
5. Measure and record the magnetic field (center frequency). This is "Frequency 1".

Note

The "Frequency 1" measurement should be taken as soon after ramping as possible.

IMPORTANT !!!

Do not move the probe between the first and second measurements. Do not move or rearrange any articles or equipment in or near the exam room during the test.

6. After 12 hours repeat Steps 1 - 5. Measure and record this as "Frequency 2".
7. Verify that the magnet has stabilized (2 Hz/hr maximum drift) by using this formula:

Drift Rate (ppm/hr)	=	$\frac{(\text{Freq 1} - \text{Freq 2}) \times 10^6}{(\text{Freq 1}) \times (12)}$
For the Initial Drift Rate use:	Freq 1 =	Initial reading
	Freq 2 =	Reading after 12 hours

9-3 PREPARATIONS FOR SHIMMING (continued)

9-3-4 Magnet Monitor



Shimming is affected by magnet pressure. Make sure the magnet monitor pressure controller is functioning properly and that the magnet pressure is steady at 3.9 - 4.1 psig before starting shimming procedure.

1. Check magnet pressure. Make sure it is maintained with the 3.9 - 4.1 psig range when taking field maps.
2. Make sure the helium level is at least 80%.

9-3-5 Quench Supercon Shield

1. Turn the Cryocooler off for 10 minutes. This will quench the Supercon Shield around the Coldhead.

Note

This only needs to be done before the initial S/C shimming and not between each shimming iteration.

9-3-6 Field Camera Set-Up Using ShimTool

Note

These instructions are for the GE ShimTool Software, Version 2.5a, (5102783) included in LCC300 Shim Camera Kit 2386028.

1. Connect the Shim Camera to the PC's COM port. Make sure all connections are secure.
2. Install the Mapping Fixture's Probe Array in conformance with Section 5-4-2, if not already installed.
3. Insert the GE ShimTool Software CD (5102783) into the computer. The CD should automatically install all ShimTool files at **C:\MRshim**.
4. Create **C:\MRshim\W000##0** directory for this magnet's ".map", ".cur" and ".shm" files. Substitute the magnet's serial number for the ##.
5. Copy the magnet's ATR shim file from the ATR floppy to the PC to the **C:\MRshim\W000##0** directory.
6. Double-click on the "Corona ShimTool" icon or the "shimtool.exe" listing in the ShimTool directory in Windows Explorer.

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-6 Field Camera Set-Up Using ShimTool (continued)**

7. When the screen shown in Illustration 9-6 appears:

- a. Verify that "W" appears beside "Series:".
- b. Type "W" into the box beside "Test Cell:".

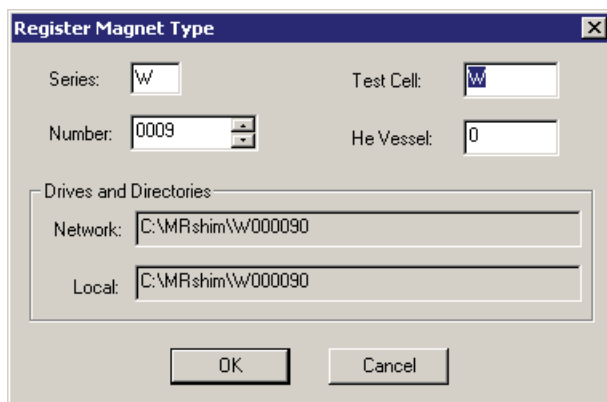
Note

Use "W" for all shimming done on a customer' site.

- c. Adjust the number displayed beside "Number:" to the serial number of the magnet to be shimmed.
- d. Finally, press <OK>.

Note

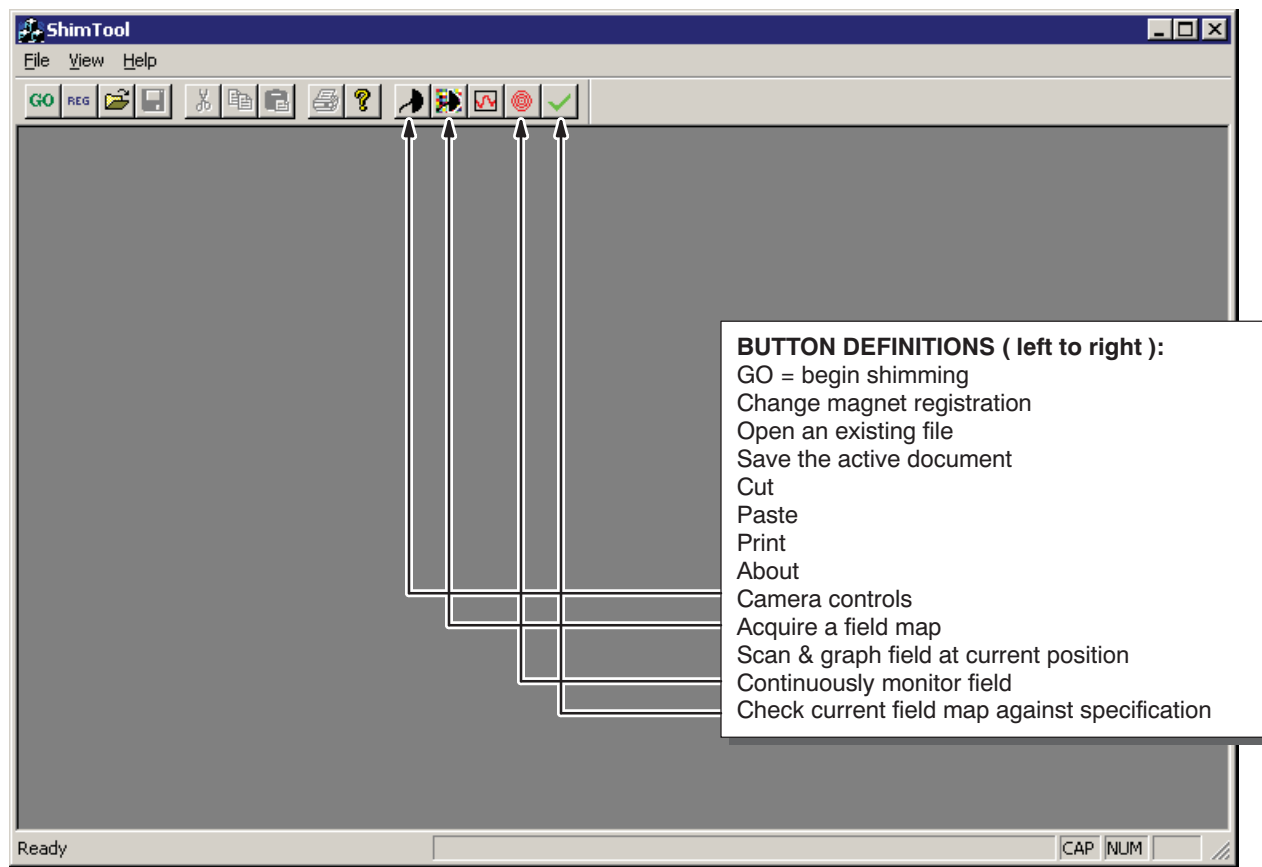
If a subdirectory for storing all ".map", ".cur" and ".shm" files was not manually created in Step 4, ShimTool will be created automatically at **C:\MRshim\W000##0** (## represents the serial number selected).



THE "REGISTER MAGNET TYPE" SCREEN
ILLUSTRATION 9-6

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-6 Field Camera Set-Up Using ShimTool (continued)**

8. Select the **[Camera Controls]** button (fifth button from right) when the screen shown in Illustration 9-7 appears.



SHIMTOOL MAIN SCREEN BUTTON DEFINITIONS
ILLUSTRATION 9-7

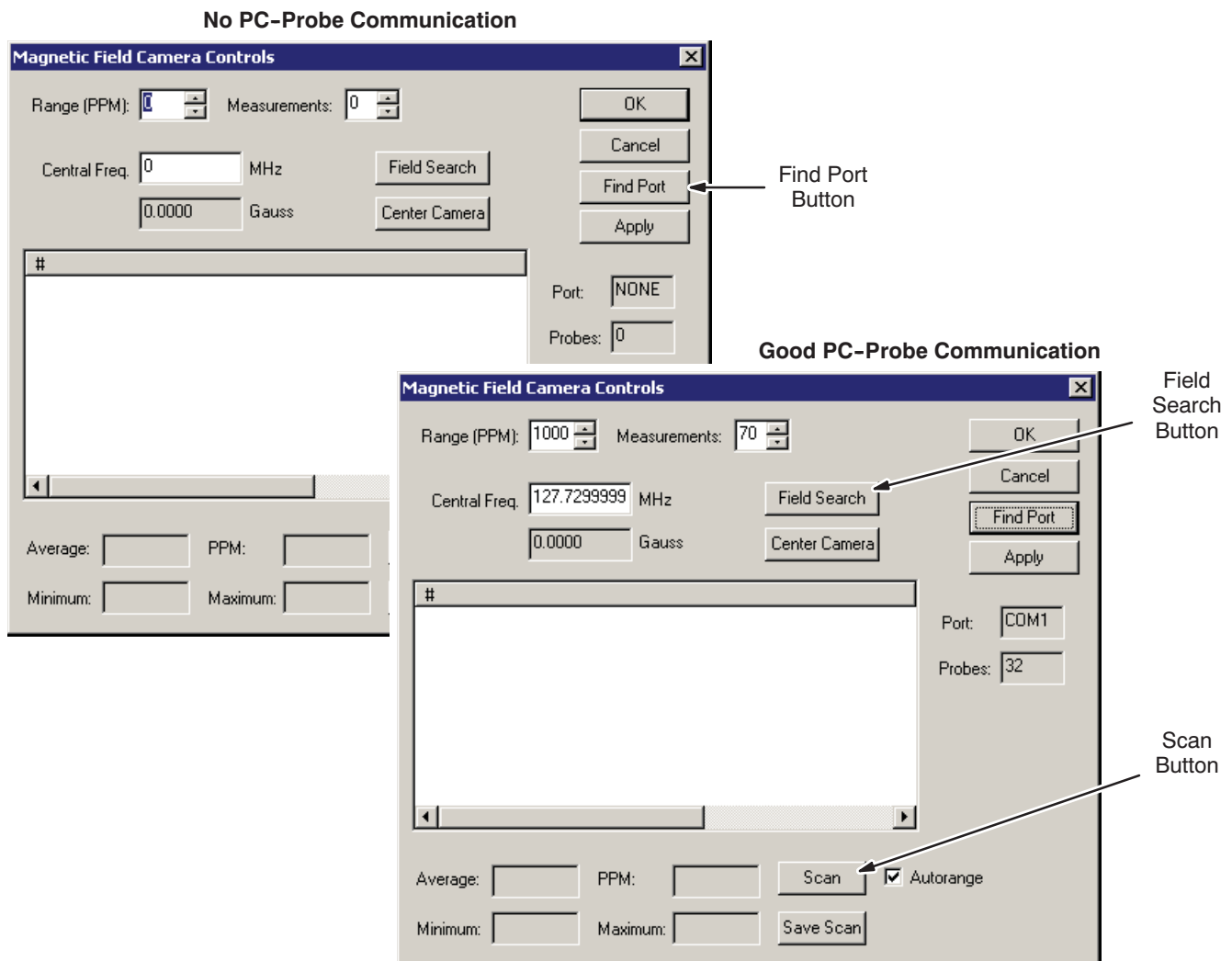
9-3 PREPARATIONS FOR SHIMMING (continued)

9-3-6 Field Camera Set-Up Using ShimTool (continued)

- Verify that the “Magnetic Field Camera Controls” window that appears (Illustration 9-8) displays a “Center Freq:” of 127.7 MHz (rounded).

Note

If the program did not find a camera attached to the PC's COM port (as in the upper left example in Illustration 9-8), verify that all connections between the probe array, camera and PC are secure, then select the **[Find Port]** button to reestablish camera-PC communications. Also press **<CTRL> + <ALT> + ** to make sure no other tasks are running on the PC besides ShimTool and Windows Explorer.



MAGNET FIELD CAMERA CONTROLS: PC-PROBE COMMUNICATION
ILLUSTRATION 9-8

9-3 PREPARATIONS FOR SHIMMING (continued)**9-3-6 Field Camera Set-Up Using ShimTool (continued)**

10. Choose an option on the “Magnetic Field Camera Controls” screen:

- Select the **[Field Search]** button to automatically find the center frequency of the magnet and update the “Center Freq.” displayed. See Illustration 9-9. An initial listing of (left to right) frequency in MHz, field strength in Gauss, standard deviation in Gauss and measurement count readings from each probe in the array is also displayed in the text box.

Note

The measurement counts should all read close to the number appearing beside “Measurements:” at the top of the screen.

- Select the **[Scan]** button to verify probe functioning by updating the scrolling text box.

11. Select **[OK]** to finish camera set-up and close the “Magnetic Field Camera Controls” screen.

Magnetic Field Camera Controls

Range (PPM): 1049 Measurements: 70

Central Freq. 127.7306316 MHz

0.0000 Gauss

Field Search

Center Camera

OK

Cancel

Find Port

Apply

#	Freq [MHz]	Field [G]	St...	R...
00	127.6747690	29987.56785	1.20	070
01	127.6754105	29987.71852	1.20	070
02	127.6762736	29987.92124	1.20	070
03	127.6766018	29987.99833	2.40	065
04	127.6763190	29987.93191	2.40	062
05	127.6761838	29987.90015	2.40	048
06	127.6770099	29988.09418	3.60	041
07	127.6789044	29988.53915	4.90	050
08	127.6810483	29989.04270	2.40	070
09	127.6827498	29989.44234	1.20	070
10	127.6847966	29989.92308	1.20	070

Port: COM1

Probes: 32

Average: 127.72872 PPM: 874.708 Scan ☒ Autorange

Minimum: 127.67476 Maximum: 127.78649 Save Scan

MAGNET FIELD CAMERA CONTROLS: AFTER CHOOSING FIELD SEARCH BUTTON
ILLUSTRATION 9-9

9-4 RE-ESTABLISHING ATR S/C SHIM CURRENTS

WARNING!

- **MAKE SURE THE MAGNET ROOM EXHAUST VENT FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM.**
- **REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.**

IMPORTANT !!!

Make sure that the shim power supply has been calibrated in conformance with the vendor's manual and that the shim heater currents are set as below before use.

Transverse 1 Shim Heater Current	610 ± 5 mA
Transverse 2 Shim Heater Current	610 ± 5 mA
Axial Shim Heater Current	710 ± 10 mA

Note

Always check ATR for any possible magnet-specific current settings.

9-4-1 Introduction

The "Recommended Total Current" values found on the Acceptance Test Report (ATR) that shipped with the magnet must be entered into the Superconducting (S/C) Correction Coils before performing field mapping and any further S/C or passive shimming.

The coil groups are:

- Transverse Coils, Odd: T1-1 through T1-5
- Transverse Coils, Even: T2-1 through T2-5
- Axial Coils: AX1 through AX4

Each coil group is connected to one heater circuit on the shim power supply. Therefore, if the only current change was in the Axial 2 Coil, turning the Axial Switch Heater on would dump the currents in Axial Coils (1 - 4) and the appropriate currents for each Axial Coil would need entering.

9-4 RE-ESTABLISHING ATR S/C SHIM CURRENTS (continued)

9-4-2 Entering ATR S/C Shim Coil Power Supply Currents

WARNING!

BEFORE TURNING ON THE SHIM COIL POWER SUPPLY AND APPLYING SHIM CURRENTS, MAKE SURE THE FOLLOWING CONDITIONS EXIST TO PREVENT IRREPARABLE DAMAGE TO THE VAPOR-COOLED SHIM LEADS:

- THE ORIFICE ON THE SHIM LEAD VENT IS IN PLACE.
- THE SHIM LEAD EXTENSION IS FROSTED AND VENTING.
- THE POWER SUPPLY CURRENT CONTROLS ARE SET TO ZERO (FULL CCW).

CAUTION

Whenever the Shim Lead is to be engaged, the Nupro Valve needs to be opened before and during engagement to avoid a possible quench. Wait for the valve to frost, then slowly engage the Shim Lead. Do not remove the orifice cap on the Shim Lead that controls the gaseous helium flow. See SET-UP AND CALIBRATION, Section 6, Shim Lead Engage / Disengage.

1. Make sure vapor is escaping from the Shim Lead and the lead is frosted, then turn on the power supply.
2. Repeat the following steps (a - g) to input ATR shim currents for each shim group in sequence (T1, then T2 and finally Axial). See Table 9-1.
 - a. Set the Shim Group Select Switch to the appropriate group (T1, T2 or Axial). See Illustration 9-10.
 - b. Set the Course / Fine Current Adjust potentiometers on the Shim Power Supply to zero (0) amps (full CCW).

Note

The polarity relays will not change polarity if there is any current flowing through their respective channels. The current must be reduced to zero before the relays will change to the position indicated by the external switches on the power supply.

9-4 RE-ESTABLISHING ATR S/C SHIM CURRENTS (continued)**9-4-2 Entering ATR S/C Shim Coil Power Supply Currents (continued)**

2. (continued)

TABLE 9-1
S/C SHIM GROUPS

Course / Fine Adjust Potentiometer & Polarity Light / Switch	'Shim Group Select' Switch Setting		
	Transverse 1 Shim Group (Middle Switch Position)	Transverse 2 Shim Group (Bottom Switch Position)	Axial Shim Group (Top Switch Position)
1	T1-1 (ZX^2Y^2)	T2-1 (ZXY)	AX1
2	T1-2 (X)	T2-2 (Y)	AX2
3	T1-3 (Z^2X)	T2-3 (Z^2Y)	AX3
4	T1-4 (ZX)	T2-4 (ZY)	AX4
5	T1-5 (X^2Y^2)	T2-5 (XY)	—
6	—	—	—

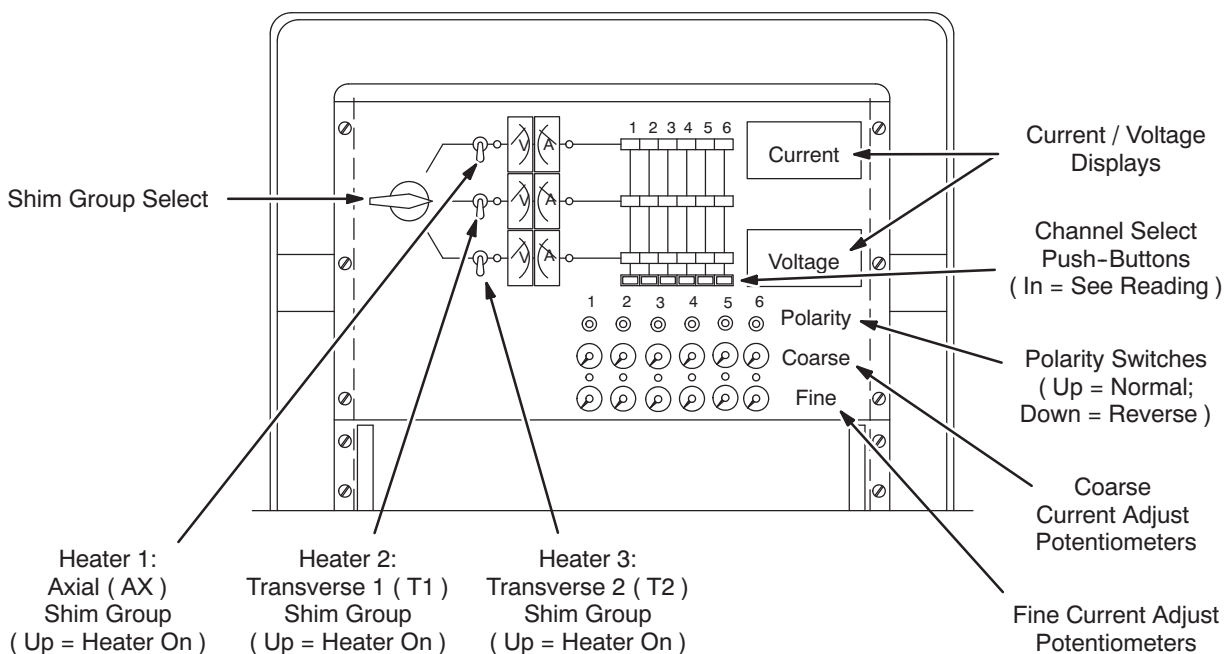
**SHIM POWER SUPPLY CONTROLS (FRONT PANEL)**

ILLUSTRATION 9-10

9-4 RE-ESTABLISHING ATR S/C SHIM CURRENTS (continued)**9-4-2 Entering ATR S/C Shim Coil Power Supply Currents (continued)**

2. (continued)



Magnet damage is possible if heater currents are not set correctly.

- c. Turn on the selected group's Switch Heater (Transverse 1, Transverse 2 or Axial) and wait 30 seconds for the switches to go resistive.

IMPORTANT !!!

On a reverse polarity magnet enter negative (-) polarity currents where the ATR says positive (+) and positive (+) polarity currents where the ATR says negative (-).

- d. Set the ATR "total" currents and polarities for the shim coil group.

IMPORTANT !!!

The LCC300 magnet correction coils have greater inductances than those of a 1.5T magnet. Therefore, a longer settling time is required for the shim currents. It is important that the shim current settling time is in conformance with Table 9-2 to prevent subsequent current drift after settling.

- e. Allow the shim current to settle for the time shown in Table 9-2.

TABLE 9-2
SHIM CURRENT SETTLING TIMES (RE-ESTABLISHING ATR)

Shim Coil Group	Settling Time
T1	20 minutes
T2	
Axial*	45 minutes
* Axial times represent minimum times. Axial coupling may require longer wait times to meet the field stabilization requirements stated in Step 3 below.	

- f. Verify the ATR currents, then turn off the heater group and wait 5 minutes.
- g. After waiting 5 minutes, gradually turn the power supply Current Adjust Potentiometers down to zero.

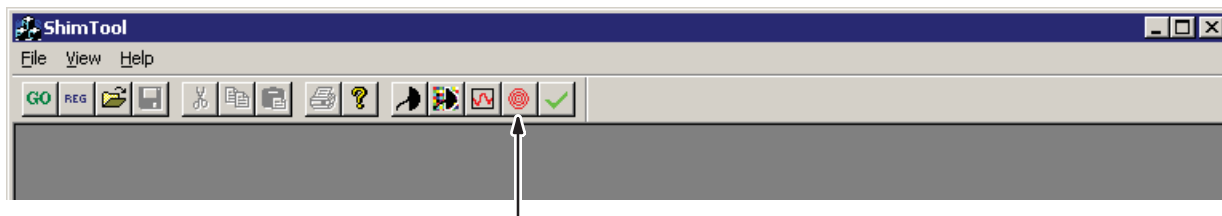
IMPORTANT !!!

Failure to accurately set the power supply currents and polarity can greatly increase the number of plotting iterations (i.e., time to shim the magnet).

9-4 RE-ESTABLISHING ATR S/C SHIM CURRENTS (continued)

9-4-2 Entering ATR S/C Shim Coil Power Supply Currents (continued)

3. For axial currents (**ONLY**), monitor field drift until it stabilizes (i.e., < 2 Hz change in total field over a two minute period):
 - a. Choose the **[Continuously Monitor Field]** button (second button from right) from ShimTool's main screen. See Illustration 9-11.



Continuously Monitor Field
Button

SHIMTOOL MAIN SCREEN BUTTONS

ILLUSTRATION 9-11

- b. Make sure the probe array is in the 30° position (slice / arc 2 in Table 9-3), then select "Probe 22."
 - c. Select **[Start]** on the screen that appears to beginning measuring field strength and drift. Make note of the starting frequency for use in step 3d.
 - d. Wait 2 minutes, then select **[Stop]** and record the change between the starting and present frequencies.
 - If the change is < 2 Hz, continue with step 4.
 - If the change is $\geq$ 2 Hz, repeat steps 3c and 3d.
4. Once the field is stable and the Axial settling time stated in Table 9-2 has elapsed, turn off the Axial Switch Heater and allow the Heater to cool for 5 minutes.
5. Slowly turn all the Course / Fine Current Adjust potentiometers on the shim power supply back down to zero (0) amps (full CCW).
6. Close the Nupro valve on the shim lead.

9-5 RUNNING SHIMTOOL

9-5-1 Field Map Acquisition

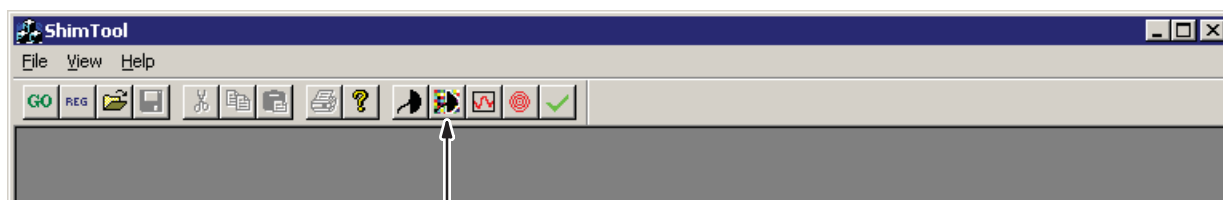
IMPORTANT !!!

The magnet's center frequency must be stable in conformance with Section 9-3-3, Field Stability Check, before any alteration to the magnet's S/C or passive shims are made.

Note

The Mapping Fixture and Shim Camera must be centered in conformance with Section 5-4-2.

1. Choose the **[Acquire Field Map]** button (fourth button from right) from ShimTool's main screen. See Illustration 9-12.



Acquire Field Map
Button

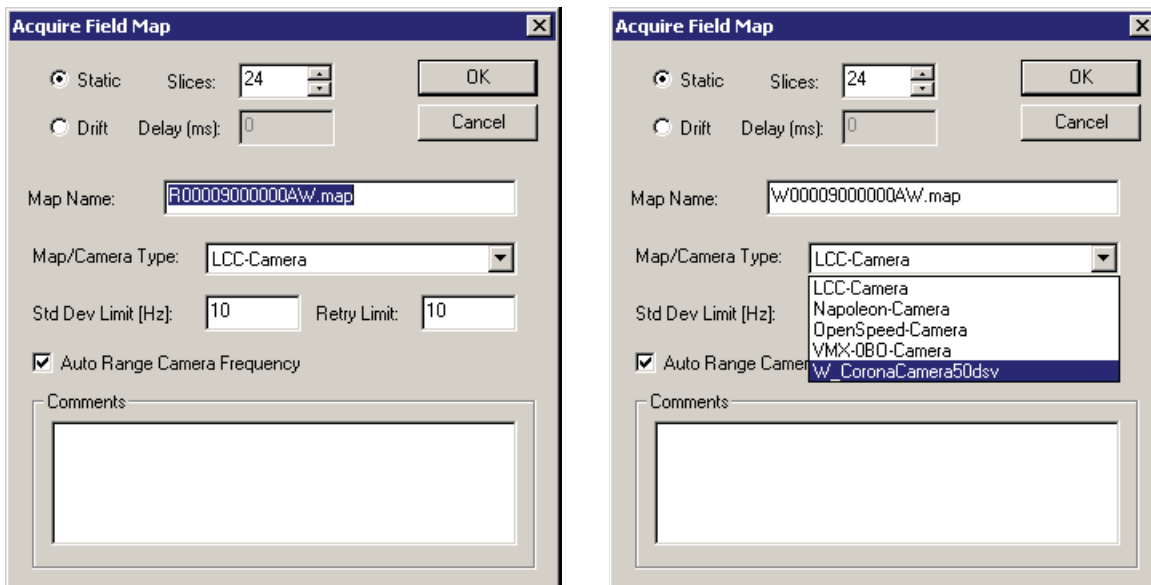
SHIMTOOL MAIN SCREEN BUTTONS
ILLUSTRATION 9-12

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

2. Select the **[Static]** button when the screen shown in Illustration 9-13 appears. Then select “W_CoronaCamera50dsv” from the “Map/Camera Type” drop-down menu.

Note

Leave the default number beside “Slices:” unchanged.



ACQUIRE FIELD MAP
ILLUSTRATION 9-13

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

3. Make sure the curved part of the field camera's probe array faces directly up (0° position).



Moving articles or equipment in the magnet room may affect field readings.

IMPORTANT !!!

Do not allow non-essential conversation / distractions during map acquisition. If an error occurs during field map acquisition:

- a. Click the **[Abort]** button in the "Acquire Progress" window on screen (Illustration 9-14) and press the field camera remote box's switch one more time.
 - b. Delete the incomplete ".map" file, but **DO NOT** delete other files.
 - c. Retake the map starting at Step 1.
4. Select **[OK]** to begin mapping the magnet's field.

Note

The field map screen will appear with the "Acquire Progress" window overlaid. See Illustration 9-14. The field camera's remote box should also start to repeatedly flash.

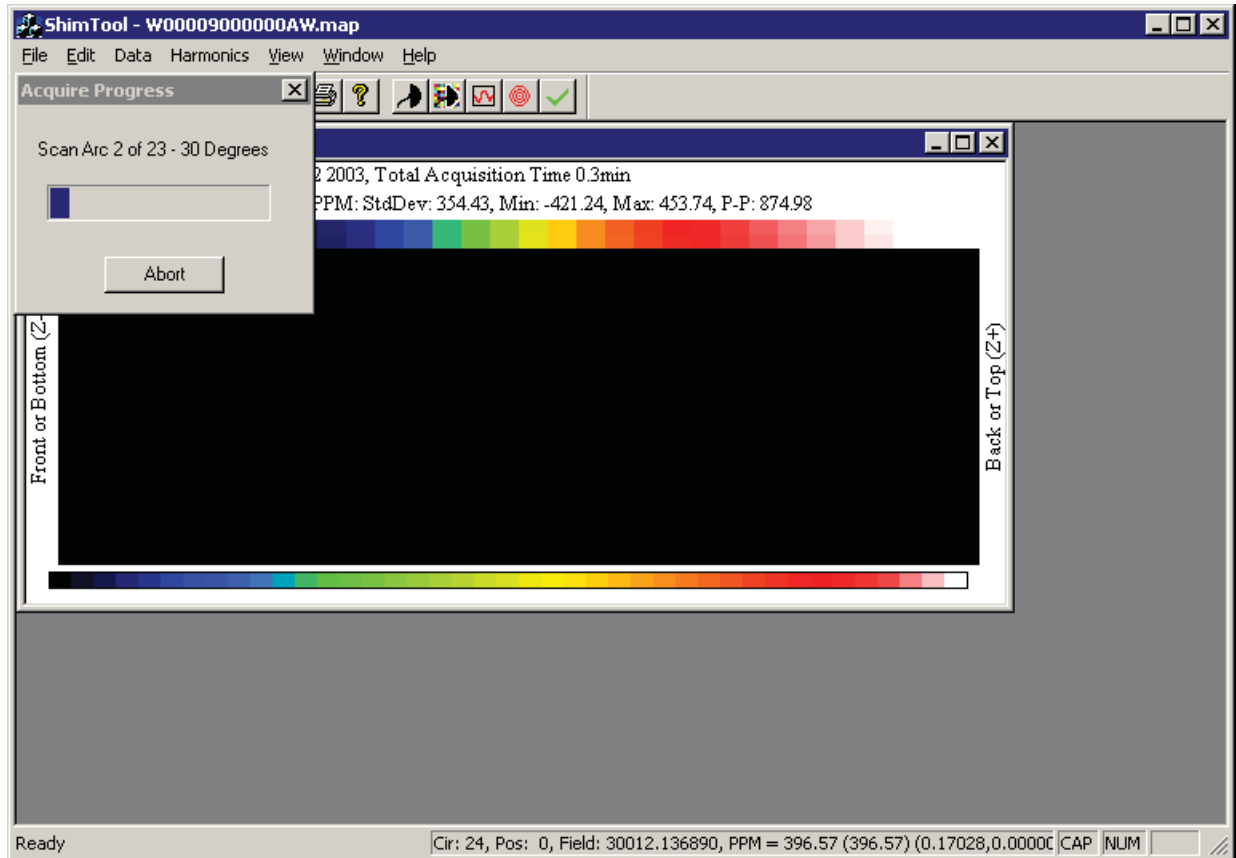
9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)****FIELD MAP BEING ACQUIRED**

ILLUSTRATION 9-14

5. Press the field camera remote box's switch after it has flashed continuously for at least 3 flashes. The light will stop flashing and stay on while one slice of the field map will be acquired (except after the last slice, when the light will go off).
6. Slowly and smoothly rotate the camera one position (15°) clockwise when the camera's push-button switch starts flashing repeatedly again.

IMPORTANT !!!

If a standard deviation error occurs while the probe array is taking a slice, the camera will automatically re-scan that slice and the remote box will briefly flash. Make sure the light has flashed continuously for at least 3 flashes before changing the camera's position.

7. Repeat steps 5 and 6 until 24 map slices (slice 0 at $0^\circ / 360^\circ$ through slice 23 at 345°) have been taken. See Table 9-3 for a summary of the slices.

Note

The remote box will flash slower at positions 0 ($0^\circ / 360^\circ$), 6 (90°), 12 (180°) and 18 (270°).

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

TABLE 9-3
FIELD CAMERA MAPPING SLICES (POSITIONS)

Slice / Arc* (Position)	Degrees	Slice / Arc* (Position)	Degrees	Slice / Arc* (Position)	Degrees	Slice / Arc* (Position)	Degrees
0	0 / 360**	6	90**	12	180**	18	270**
1	15	7	105	13	195	19	285
2	30	8	120	14	210	20	300
3	45	9	135	15	225	21	315
4	60	10	150	16	240	22	330
5	75	11	165	17	255	23	345

* The slice / arc number shown here and in ShimTool corresponds with the number stamped on the mapping fixture's positioning wheel.

** The remote box flashes slower at these positions.

Note

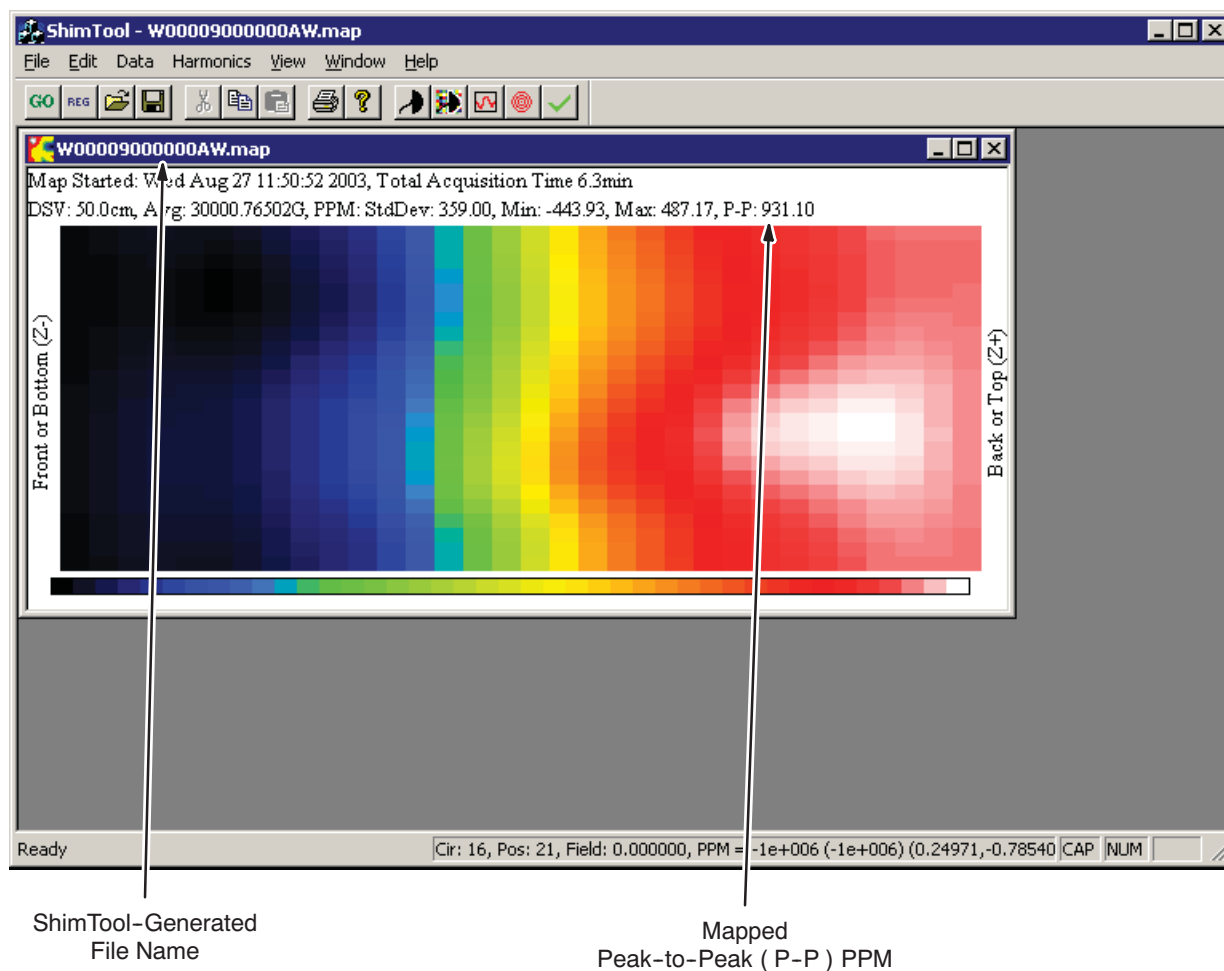
When all 24 slices have been taken, the finished map appears on the field map screen (Illustration 9-15). ShimTool automatically assigns a sequential alphabetic name to both the displayed map and the ".map" file it creates. The file name consists of:

- The magnet series and number information entered on the "Register Magnet Type" screen in Section 9-3-6, steps 7a and 7c, respectively. ("W" + "00009" becomes "W00009".)
- A sequential alphabetic character. ("A", "B," ... "C" becomes "000000A", "000000B", ... "000000C".)
- The test cell entered on the "Register Magnet Type" screen in Section 9-3-6, step 7b.
- Plus the ".map" file extension.

9-5 RUNNING SHIMTOOL (continued)

9-5-1 Field Map Acquisition (continued)

8. Record the mapped peak-to-peak (P-P) ppm figure displayed on the field map screen (Illustration 9-15) in DATA SHEETS, Table 6-1, on the same line as the map name.

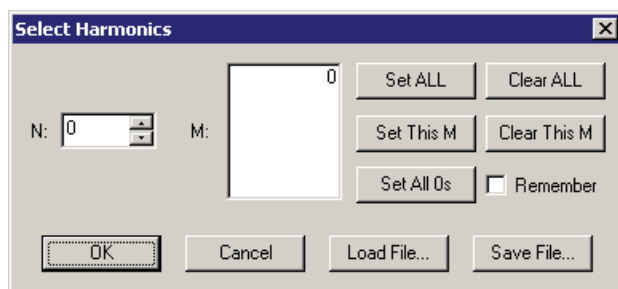


SAMPLE FIRST FIELD MAP SCREEN (FOR AXIAL CENTERING)
ILLUSTRATION 9-15

9. If this is the first field map taken since the mapping fixture was installed or moved in any substantial way, then use this map to axially center the probe array:
 - a. Choose **“Decompose (F6)”** from the **“Harmonics”** menu (fourth menu from the left in Illustration 9-15).
 - b. Select **[OK]** when the screen in Illustration 9-16 appears.

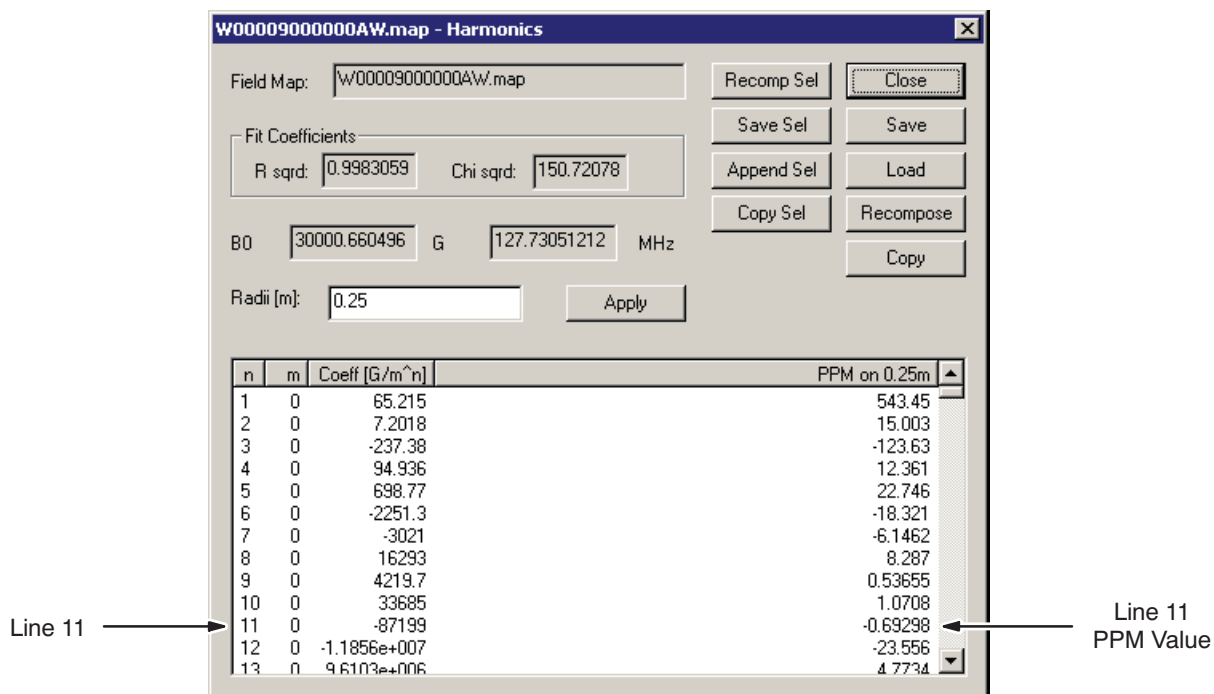
9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

9. (continued)



SELECT HARMONICS
ILLUSTRATION 9-16

- c. Look at the PPM value displayed at the right end of line 11 when the screen in Illustration 9-17 appears.
- d. If the PPM value on line 11 (harmonic 11, 0) is > -1.0 ppm and $< +1.0$ ppm, the probe array is centered axially. Continue with step 10, and use this 'centered map' to run ShimTool set for S/C shimming.



FIELD MAP HARMONICS
ILLUSTRATION 9-17

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

9. (continued)

e. Divide this PPM value by 1.7.

$$\frac{\text{PPM Value on Line 11}}{1.7} = \frac{\text{Number of Turns}}{\text{of Mapping Fixture's Axial Positioning Screw Required to Axially Center Probe Array}}$$

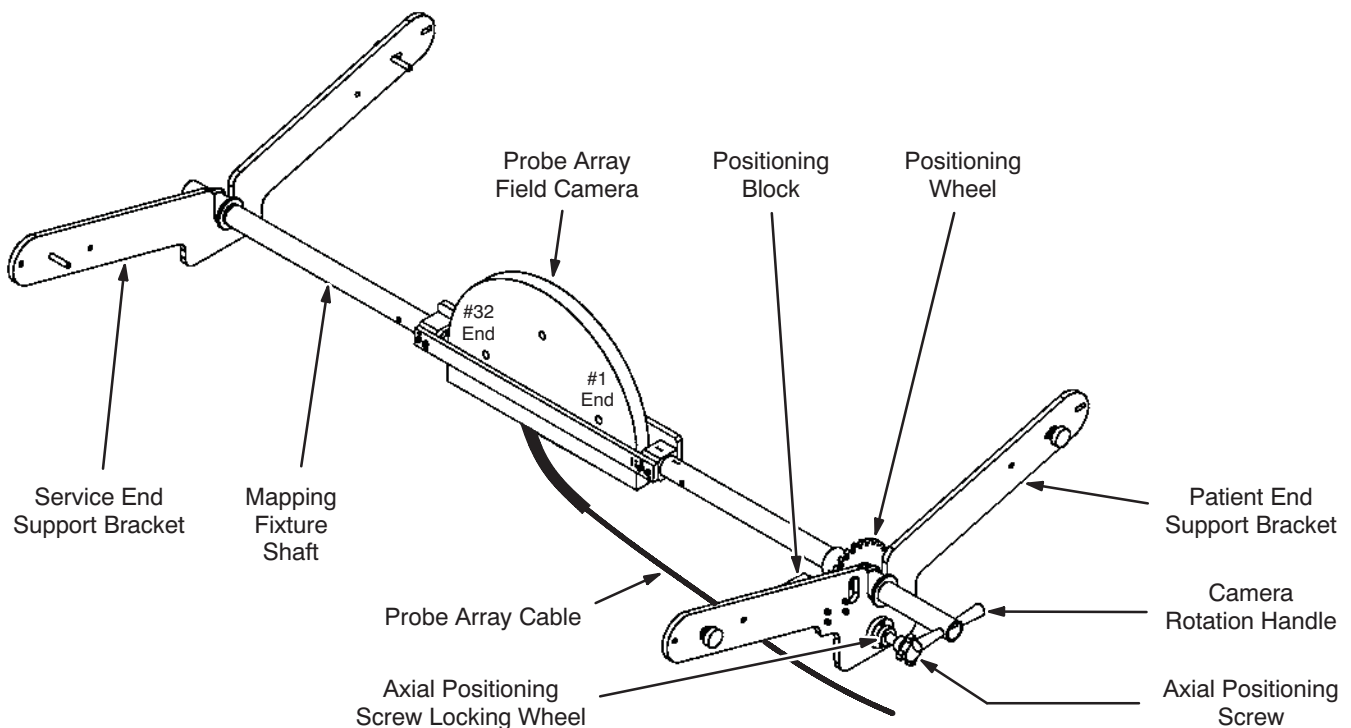
Note

Positive (+) PPM value: turn clockwise (CW).

Negative (-) PPM value: turn counterclockwise (CCW).

Turning in increments smaller than a quarter turn is not necessary.

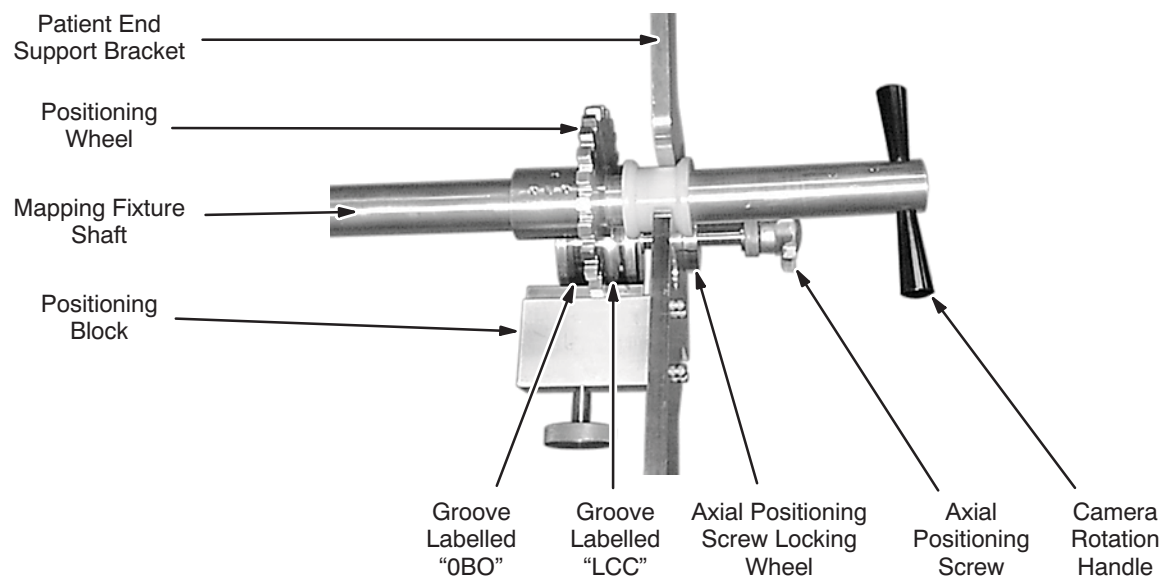
f. Loosen the axial positioning screw locking wheel, carefully turn the axial positioning screw (see Illustrations 9-18 and 9-19) the number of full turns calculated in step 9e and then tighten the locking wheel.



LCC300 FIELD MAPPING FIXTURE 2380758 BASIC COMPONENTS
ILLUSTRATION 9-18

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

9. (continued)

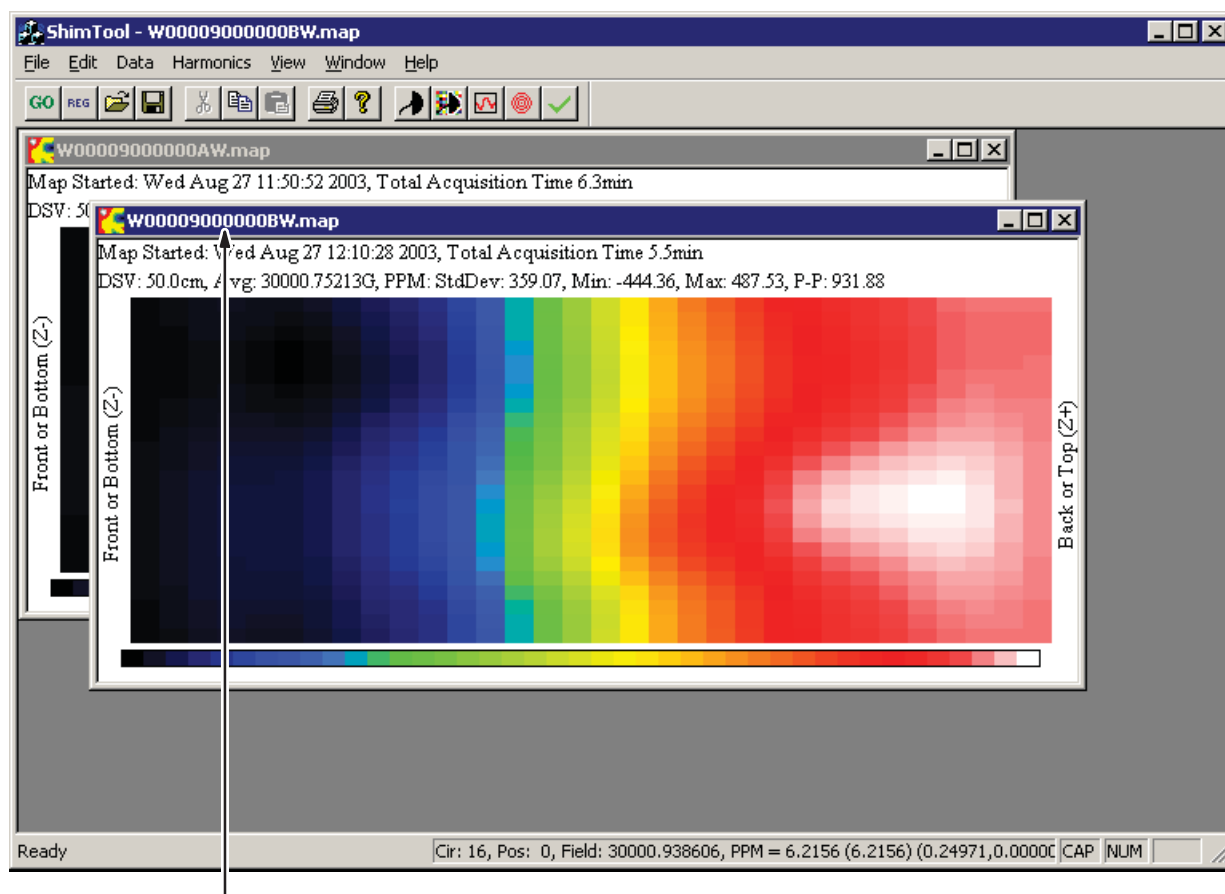


MAPPING FIXTURE'S AXIAL POSITIONING SCREW
ILLUSTRATION 9-19

- g. Record the ShimTool-generated ".map" file name in DATA SHEETS, Section 6, Shimming Process Data, in the upper section ("Maps for Probe Centering").
- h. Repeat steps 1 - 7 and 9a - 9d to verify axial centering of the probe array.

9-5 RUNNING SHIMTOOL (continued)**9-5-1 Field Map Acquisition (continued)**

10. Record the ShimTool-generated “.map” file name in the appropriate DATA SHEETS section(s):
 - Record map names used for passive shimming in DATA SHEETS, Sections 6 and 7.
 - Record map names used for superconducting shimming in DATA SHEETS, Sections 6 and 8.
11. Close ShimTool’s harmonics window(s).



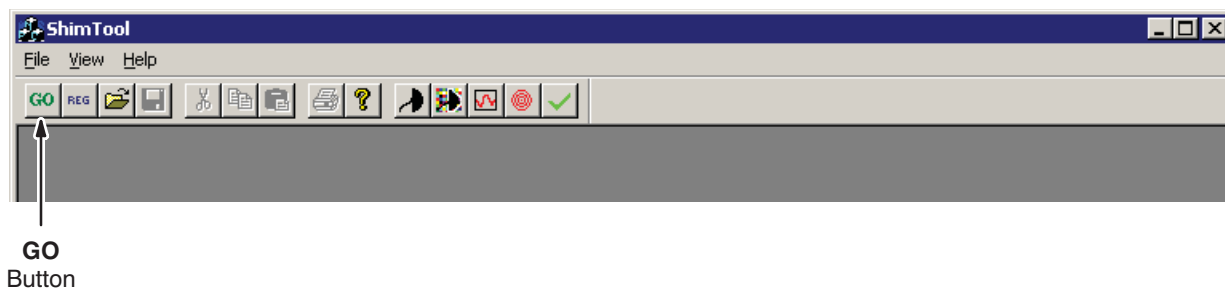
ShimTool-Generated
File Name

SAMPLE FIELD MAP SCREEN FOR SHIMMING
ILLUSTRATION 9-20

9-5 RUNNING SHIMTOOL (continued)

9-5-2 Acquiring a Shim File

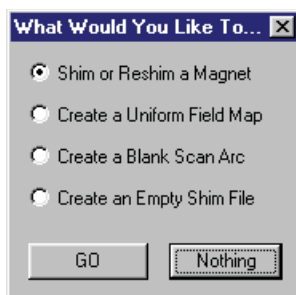
1. Choose the **[GO]** button (first button on left) from ShimTool's main screen. See Illustration 9-21.



SHIMTOOL MAIN SCREEN BUTTONS

ILLUSTRATION 9-21

2. Select **[Shim or Reshim a Magnet]** and then **[GO]** when the screen shown in Illustration 9-22 appears.



THE "WHAT WOULD YOU LIKE TO DO?" SCREEN

ILLUSTRATION 9-22

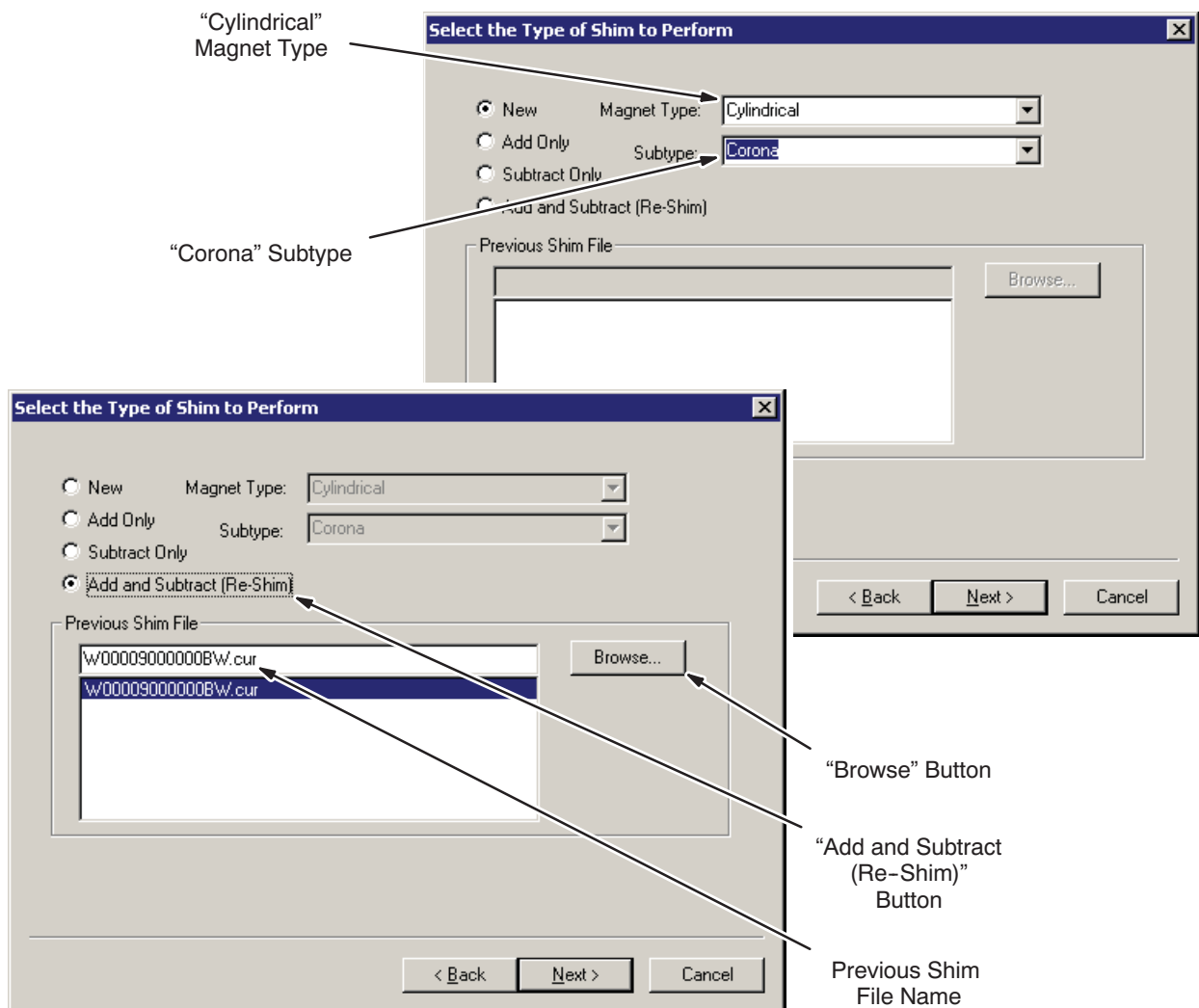
9-5 RUNNING SHIMTOOL (continued)

9-5-2 Acquiring a Shim File (continued)

- When the “Select the Type of Shim to Perform” screen shown in Illustration 9-23 appears, verify that the “Magnet Type” field reads “Cylindrical” and that the “Subtype” field reads “Corona”, then select the **[Add and Subtract (Re-Shim)]** button.

Note

Verify the “Magnet Type” and “Subtype” fields before selecting **[Add and Subtract (Re-Shim)]** since these fields cannot be changed while **[Add and Subtract (Re-Shim)]** is selected. ShimTool is only run with the **[New]** button selected at the factory.



THE “SELECT THE TYPE OF SHIM TO PERFORM” SCREEN
ILLUSTRATION 9-23

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

4. Accurately select the “Previous Shim File” name (the last file with the “.cur” extension for S/C shimming or “.shm” extension for passive shimming) from the list beside “Previous Shim File:”.

IMPORTANT !!!

Use the ATR “.cur” / “.shm” file names as “Previous Shim File” for the first iteration of Shim-Tool after the ATR S/C currents are reestablished. Make sure these “.cur” / “.shm” files have been copied to **C:\MRshim\W000##0** before starting the iteration.

IMPORTANT !!!

Do not enter incorrect file names. Keep track of filenames and iterations used — and what they were used for — by entering this information in DATA SHEETS, Sections 6 (master log of shimming process) and either 7 (passive shimming) or 8 (superconducting shimming), so that accurate filenames can be entered at any later time.

Note

The path to the S/C shim file should be “**C:\MRshim\W000##0*.cur**” (with the asterisk representing the filename). The path to the passive shim file should be “**C:\MRshim\W000##0*.shm**”. If the correct shim file is not located there and you do not remember the full path to the correct shim file, select the **[Browse]** button, locate the correct file, select it, and then open the file.

5. Verify all information on the “Select the Type of Shim to Perform” screen is accurate, then select **[Next>]**.

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)****IMPORTANT !!!**

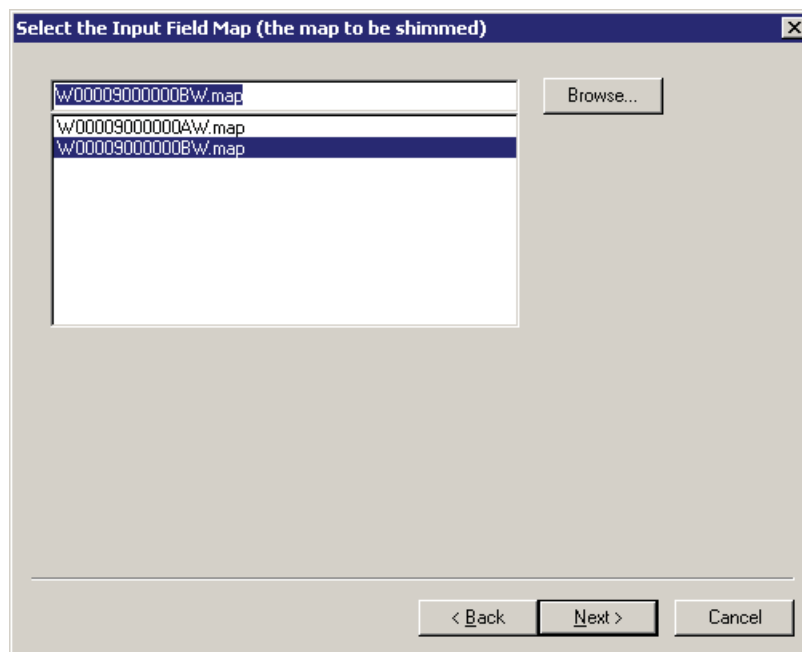
Do not enter incorrect file names. Keep track of filenames and iterations used — and what they were used for — by entering this information in DATA SHEETS, Sections 6 (master log of shimming process) and either 7 (passive shimming) or 8 (superconducting shimming), so that accurate filenames can be entered at any later time.

6. When the “Select the Input Field Map” screen shown in Illustration 9-24 appears, accurately enter the filename of the last field map (a “.map” extension) taken in Section 9-5-1, Field Map Acquisition.

Note

The path to the field map file should be “**C:\MRshim\W000##0*.map**” (with the asterisk representing the filename). If the correct map file is not located there and you do not remember the full path to the correct map file, select the **[Browse]** button, locate the correct file, select it, and then open the file.

7. Select **[Next>]**.

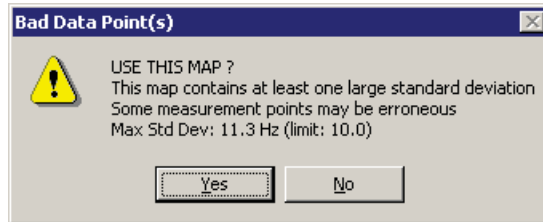


THE “SELECT THE INPUT FIELD MAP” SCREEN
ILLUSTRATION 9-24

Note

If the map file specified in step 6 contains bad data, the alert shown in Illustration 9-25 appears. Select **[No]** and then retake the map in conformance with Section 9-5-1, Field Map Acquisition, making sure all connections are good and that the cable isn’t wrapping around the mapping fixture.

If the specified map file is good, a screen shown in Illustration 9-26 should appear.

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

THE “BAD DATA POINT(S)” ALERT
ILLUSTRATION 9-25

8. Check **[Unified/Multi-Volume Shimming]** when the “Shim Target and Tolerance Controls” screen (Illustration 9-26) appears.
 - a. Make sure the “Map Type” field reads “G3UnifiedTarget”.
 - b. Select the **[Target (TOL) File]** button.
 - If the ATR was produced using ‘Tolerance Shimming,’ the path to “G3UnifiedTarget.tol” appears in the field to the right of the **[Target (TOL) File]** button.

Note

The path to the tolerance map file should be “**C:\MRshim\ShimTool\G3UnifiedTarget.tol**”. If the correct tolerance file is not located there and you do not remember the full path to the correct location, select the **[Browse]** button, locate the correct file, select it, and then open the file.

- If the ATR was produced using ‘Surface Shimming,’ a string of five numbers separated by commas appears in the field to the right of the **[Target (TOL) File]** button.

Note

The choice between Tolerance and Surface Shimming is made at the factory to give the best shimming results. This choice only effects the screen shown in Illustration 9-26, the printout shown in Illustration 9-34 and the quantity of shims used, but has **NO** procedural impact on the Field Engineer.

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

Shim Target and Tolerance Controls

☒ Unified/Multi-Volume Shimming Map Type: G3UnifiedTarget

☒ Target (TOL) File: \\MRshim\\ShimTool\\G3UnifiedTarget.to Browse...

☐ Manual Target PPM (Comma Separated)

☐ Auto-Tolerancing ☐ Critical Targeting

PPM Threshold: 0 Release Amount [PPM]: 0

Change Limit: 0

< Back Next > Cancel

**When ATR Includes
Tolerance Shimming**

Path to Tolerance File

Shim Target and Tolerance Controls

☒ Unified/Multi-Volume Shimming Map Type: G3Unified5V

☒ Target (TOL) File: 9.5,6.5,2,.3,.05 Browse...

☐ Manual Target PPM (Comma Separated)

☐ Auto-Tolerancing ☐ Critical Targeting

PPM Threshold: 0 Release Amount [PPM]: 0

Change Limit: 0

< Back Next > Cancel

**Only When
ATR Includes
Surface Shimming**

Five Surface Shim
Numbers from ATR

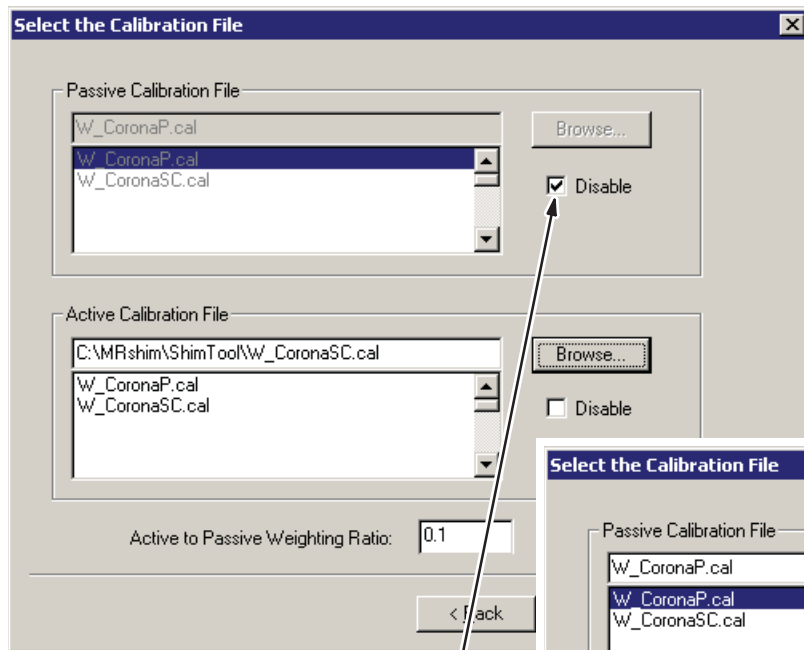
SHIM TARGET AND TOLERANCE CONTROLS

ILLUSTRATION 9-26

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

9. Select **[Next>]** to proceed to the “Select the Calibration File” screen shown in Illustration 9-27.

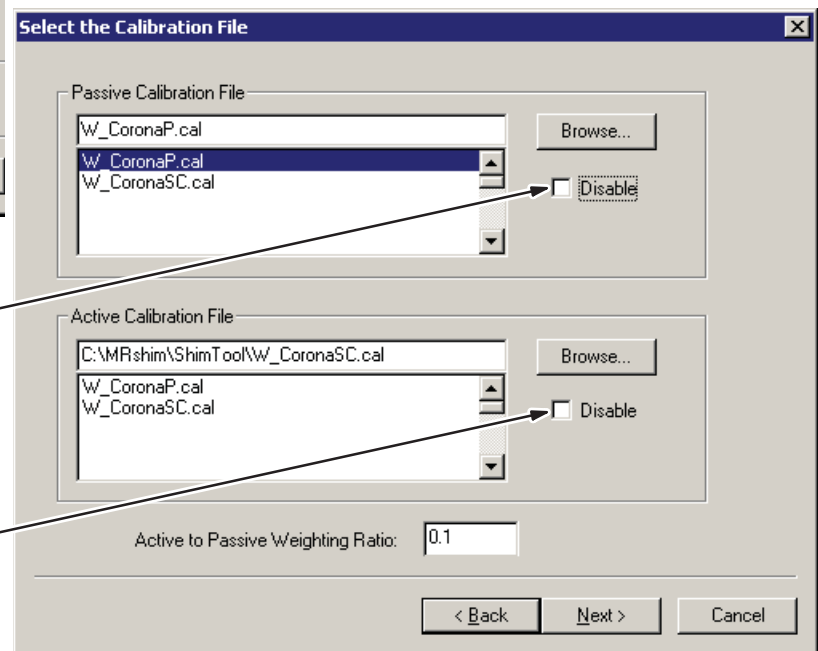
For Superconducting (Active) Shimming ONLY



For BOTH Passive & Active Shimming

Passive Shim
Calibration File
“Disable” Button

Active Shim
Calibration File
“Disable” Button



THE “SELECT THE CALIBRATION FILE” SCREEN
ILLUSTRATION 9-27

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

10. Select "W_CoronaP.cal" from the scrolling list of available calibration file list below "Passive Calibration File". See Illustration 9-27.
11. Select "W_CoronaSC.cal" from the scrolling list of available calibration file list below "Active Calibration File". See Illustration 9-27.

Note

If either the active or passive calibration files do not appear in the respective scrolling lists, select the **[Browse]** button, locate the correct file, select it, and then open the file. The paths to the calibrations files should be "C:\MRshim\Shimtool\W_CoronaP.cal" and "C:\MRshim\Shimtool\W_CoronaSC.cal", respectively.

12. If only one type of shimming (passive or active) will be done during this iteration (see Illustration 9-1, Shimming Process Flowchart), check the **[Disable]** box beside the type of shimming NOT being done.
 - Check the **[Disable]** box beside "Passive Shim Calibration File" if preparing to ONLY perform superconducting (active) shimming (Section 9-7).

Note

Disable the "Passive Shim Calibration File" when running ShimTool with the 'centered map' made in Section 9-5-1.

- Check the **[Disable]** box beside "Active Shim Calibration File" if preparing to ONLY perform passive shimming (Section 9-6). (Not recommended.)

Note

Normally, the "Active Shim Calibration File" is never disabled.

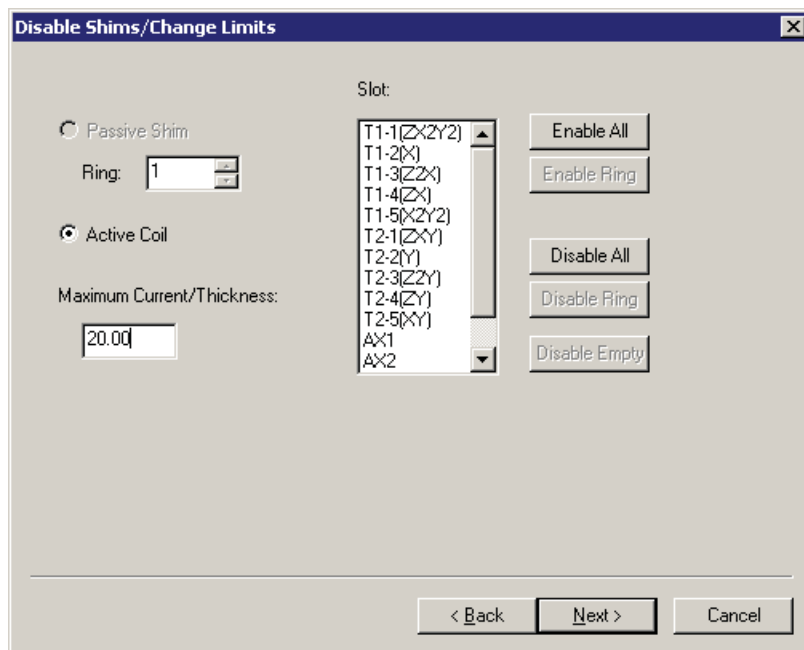
IMPORTANT !!!

ShimTool will allocate the results of its calculations to all forms of shimming that are not disabled. Extra shimming iterations may result from not setting the disable boxes accurate to the type of shimming being done.

13. Select **[Next>]**.

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

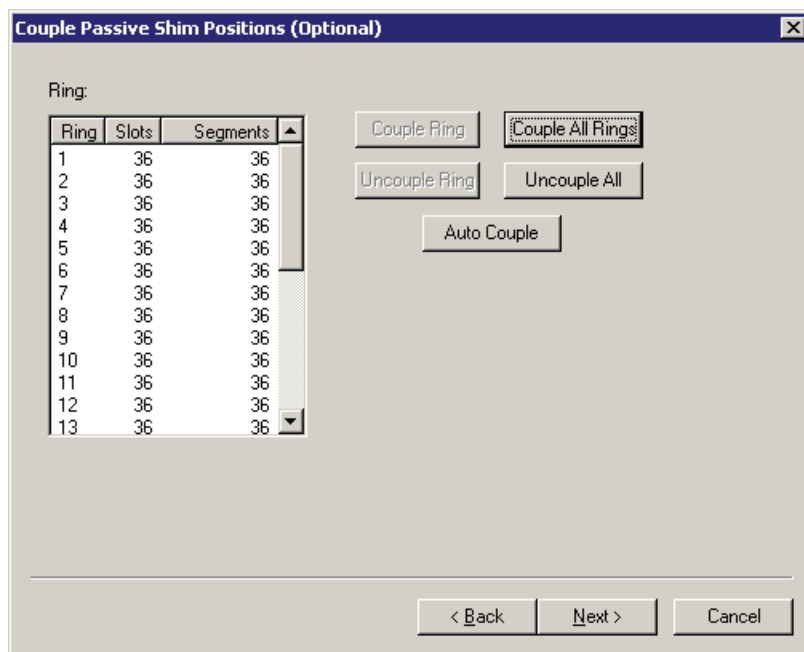
14. Select [**Next>**] without making changes to the “Disable Shims / Change Limits” screen (Illustration 9-28) when it appears.



THE “DISABLE SHIMS / CHANGE LIMITS” SCREEN
ILLUSTRATION 9-28

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

15. Select **[Next>]** without making changes to the “Couple Passive Shim Positions (Optional)” screen (Illustration 9-28) when it appears.



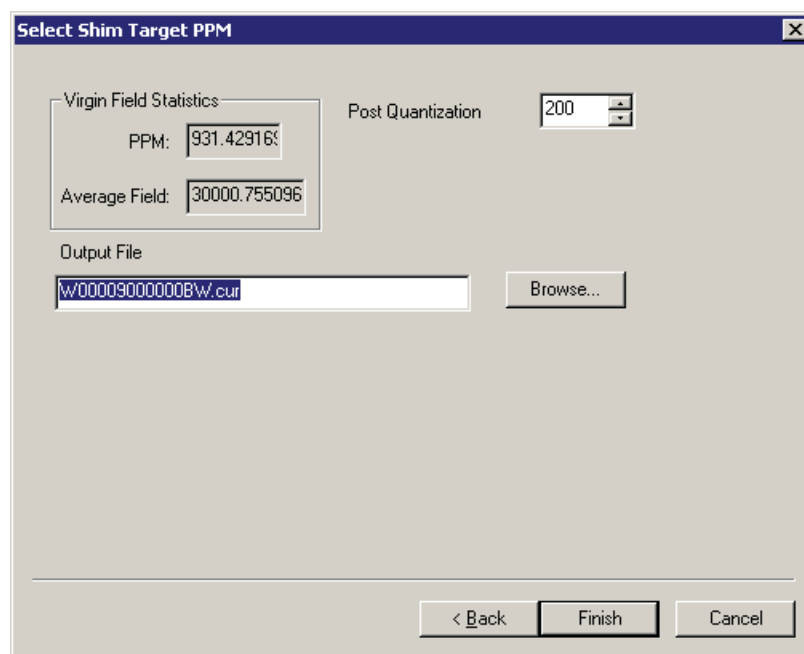
THE “COUPLE PASSIVE SHIM POSITIONS (OPTIONAL)” SCREEN
ILLUSTRATION 9-29

9-5 RUNNING SHIMTOOL (continued)**9-5-2 Acquiring a Shim File (continued)**

16. Verify that the “Output File” name on the “Select Shim Target PPM” screen (Illustration 9-30) that next appears is the next sequential alphabetical name from the input “.map” file. Make no other changes to the “Select Shim Target PPM” screen.

Note

The path to the field map file should be “C:\MRshim\W000##0*.map” (with the asterisk representing the filename). If the correct map file is not located there and you do not remember the full path to the correct map file, select the [**Browse**] button, locate the correct file, select it, and then open the file.



THE “SELECT SHIM TARGET PPM” SCREEN
ILLUSTRATION 9-30

17. Use the [**<Back**] button as required to review and verify as correct all entries and choices on previous screens.
18. Select [**Finish**]. ShimTool will begin processing the shimming information.

Note

A “Shim Status” window will appear while ShimTool is processing and disappear when the processing is complete.

9-5 RUNNING SHIMTOOL (continued)

9-5-3 After ShimTool Processing

1. Select **[Dismiss]** when the “Shim Summary” screen (like that shown in Illustration 9-31) appears.

The screenshot shows a window titled "Shim Summary - W00009000000BW.cur". It contains several input fields and a button. Arrows point from text labels to specific fields:

- "Total Shim Positions Changed (Spec.: < 5)" points to the "Changed Shims" field, which contains "0".
- "Total Shim Mass Changed (Spec.: < 5 mm)" points to the "Net Mass Change [g]" field, which contains "0.0000".

Other fields include:

- "Total Shims": 0
- "Total Mass [g]": 0.0000
- "Iterations": 61
- "Predicted Field" section:
 - "Target PPM": 9.8643
- "Unrounded" section:
 - "PPM": 52.5246
 - "Average": 30003.279
- "Quantized" section:
 - "PPM": 52.5246
 - "Average": 30003.279
- A "Dismiss" button at the bottom.

THE “SHIM SUMMARY” SCREEN
ILLUSTRATION 9-31

2. A more complete display of the shim iteration results will appear (like that shown in Illustration 9-32). To view more about the shimming results:
 - Choose “Print Lists...” from the “File” menu, set the page orientation to “Landscape” and then select the **[OK]** button to print a paper version of the multi-page ShimTool results report shown on screen in Illustration 9-32 *OR*
 - Choose “Show the output SHM file” from the “Shim” menu to save the “.cur” / “.shm” file. Open the resulting text file in NotePad, WordPad or a word processor to get a results report as in Illustration 9-33.

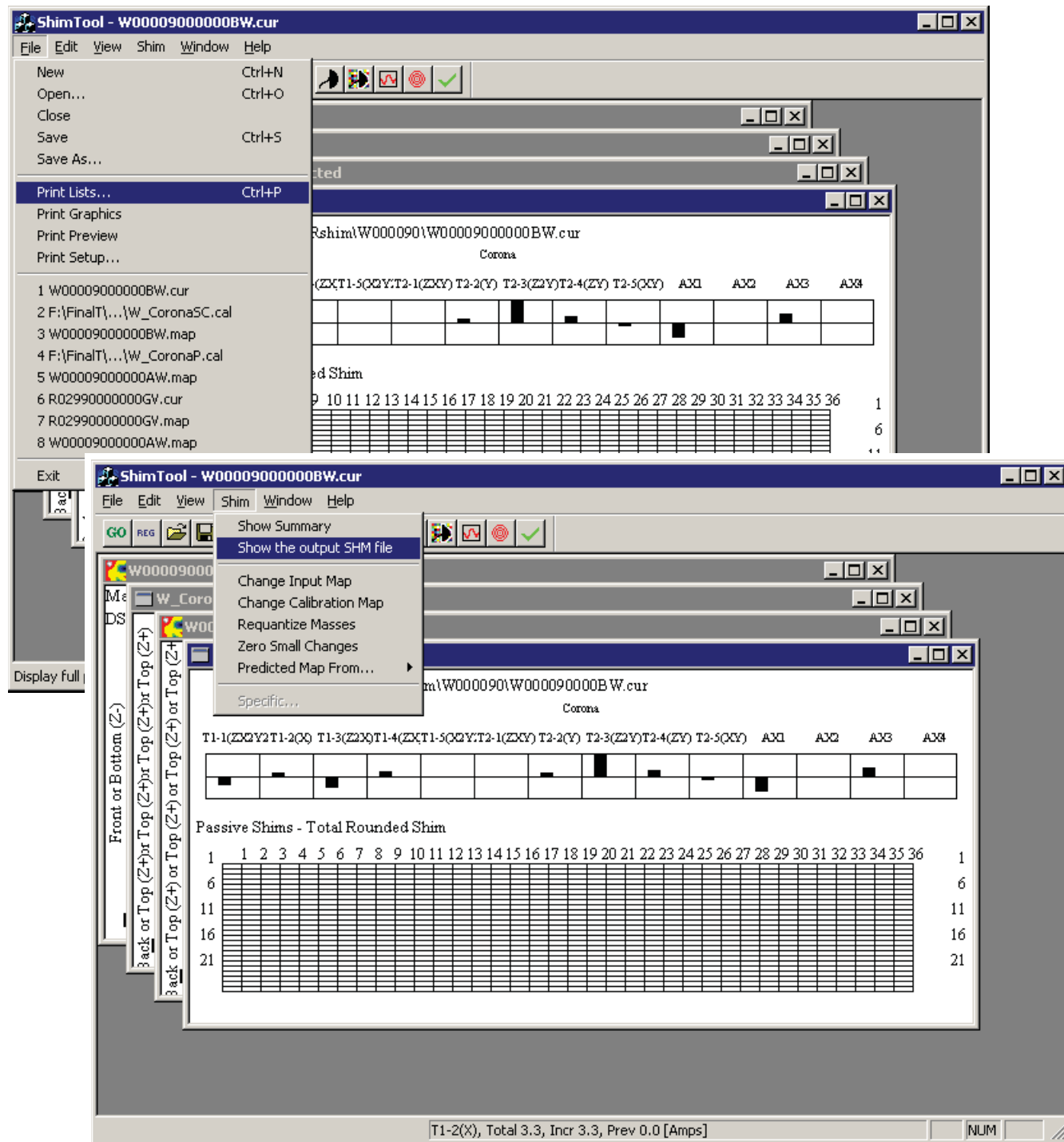
Note

The “Passive Shims – Incremental Rounded Shim” information on page 3 of ShimTool’s printout (Illustration 9-33) is the same information as contained under “Incremental – Quantized” in the “.shm” file (Illustration 9-34). (“.cur” files do not contain incremental information.)

The “Passive Shims – Total Rounded Shim” information on page 4 of ShimTool’s printout (Illustration 9-33) is the same information as contained under “Total – Quantized” in the “.shm” file (Illustration 9-34).

9-5 RUNNING SHIMTOOL (continued)

9-5-3 After ShimTool Processing (continued)



SHIM ITERATION RESULTS DISPLAY
ILLUSTRATION 9-32

9-5 RUNNING SHIMTOOL (continued)**9-5-3 After ShimTool Processing (continued)**

C:\Shimtool\W000##0\W000##000000EW.shm
Corona

Pg. 1 of 4

Shim File: W000##000000EW.shm
Magnet Type: Cylindrical - Corona
Shim Solution Time: Thu Aug 28 04:46:36 2003

Total Number of Shims Positions: 187
Changed Number of Shims Positions: **6**
Total Shim Mass: 768.0000 Shims
Net Mass Change: **2.0000 (+4.0000/-2.0000) shims**
Anticipated Field Inhomogeneity: 11.249 PPM

Input Field File: C:\MRshim\W000##0\W000##000000EW.shm
Passive Calibration File: C:\MRshim\W000##0\W_CoronaP.cal
Active Calibration File: C:\MRshim\W000##0\W_CoronaSC.cal

Shim Positions To Change (Spec.: < 5)

Net Mass Change (Spec.: < 5 mm)

C:\MRshim\W000##0\W000##000000EW.shm
Corona

Pg. 2 of 4

Active Shim Currents

Coil / Supply	Previous	Change	New
T1-1(ZX2Y2)	-6.155 A	0.000 A	-6.155 A
T1-2(X)	3.605 A	-0.024 A	3.581 A
T1-3(Z2X)	0.282 A	0.000 A	0.282 A
T1-4(ZX)	4.970 A	0.000 A	4.970 A
T1-5(X2Y2)	-1.967 A	0.026 A	-1.941 A
T2-1(ZXY)	-1.466 A	0.000 A	-1.466 A
T2-2(Y)	2.714 A	0.041 A	2.755 A
T2-3(Z2Y)	15.000 A	0.000 A	15.000 A
T2-4(ZY)	4.546 A	0.000 A	4.546 A
T2-5(XY)	-1.544 A	0.000 A	-1.544 A
AX1	-9.584 A	-0.001 A	-9.585 A
AX2	-1.024 A	0.015 A	-1.009 A
AX3	4.745 A	-0.011 A	4.734 A
AX4	0.657 A	0.008 A	0.665 A

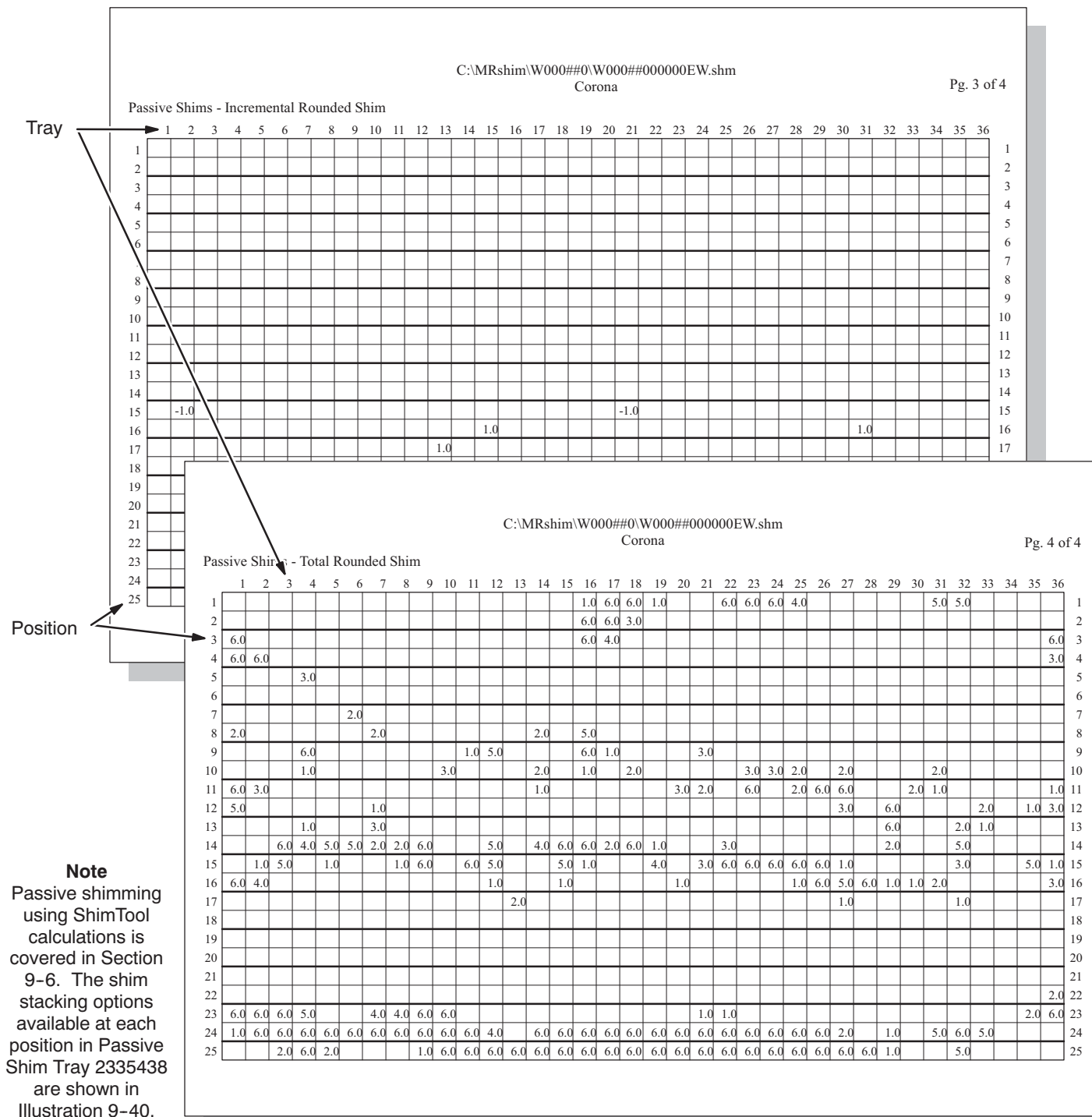
Note
Superconducting (active)
shimming using ShimTool
calculations is covered in
Section 9-7.

EXAMPLE SHIMTOOL PRINTED REPORT (PAGES 1 AND 2)

ILLUSTRATION 9-33

9-5 RUNNING SHIMTOOL (continued)

9-5-3 After ShimTool Processing (continued)



EXAMPLE SHIMTOOL PRINTED REPORT (PAGES 3 AND 4)
ILLUSTRATION 9-33 (CONTINUED)

9-5 RUNNING SHIMTOOL (continued)

9-5-3 After ShimTool Processing (continued)

“.CUR” FILE
Showing Tolerance
Shimming Report

Shim Positions To Change
(Spec.: < 5)

Net Mass Change
(Spec., S/C only: 0 mm)

Predicted PPM

Present (& Spec.) Values

48 x 45 EV Peak-to-Peak

40 cm DSV Peak-to-Peak

30 cm DSV Peak-to-Peak

20 cm DSV Peak-to-Peak

Predicted Values
(After This Iteration)

48 x 45 EV Peak-to-Peak

40 cm DSV Peak-to-Peak

30 cm DSV Peak-to-Peak

20 cm DSV Peak-to-Peak

```

SHIM OUTPUT FILE -- Generated By ShimTool

      SHIM MAGNET TYPE : Cylindrical
      MAGNET TYPE      : Corona
      AUTO-TOLERANCE   : 0.0000 @ 0.0000 PPM
      CRITICAL TARGET  : 0.0000 PPM
      INPUT FIELD FILE  : W00009000000GV.map
      OUTPUT FILE       : C:\MRshim\W000160\W00016000000GV.cur
      PREVIOUS FILE     : C:\MRshim\W000160\W00016000000FV.cur
      CALIBRATION FILE  :

ANALYSIS OPTION: ADDITION AND SUBTRACTION (RE-SHIMMING)
MINIMIZATION TECHNIQUE : ABSOLUTE

                                     ROUNDED
                                     -----
Total Number of Shim Pockets          187
Changed Number of Shim Pockets         0
Changed Number of SC Shims             13
Total Passive Shim Mass (shims)        768.0000
Net Passive Shim Mass Change (shims)    0.0000 (+0.0000/0.0000)
Anticipated Field Tolerances (PPM)     10.152

AVERAGE FIELD (Gauss)                 Initial= 29998.85
                                      Final= 29998.85

FIELD TOLERANCE (PPM)
                                      Requested= unified
                                      Final Before Rounding= 10.152

The mapped field inhomogeneity is ( 768 pts)    21.93
The reported field inhomogeneity is ( 768 pts )  21.93
NUMBER OF LP ITERATIONS                    =      44
RUN COMPLETED Thu Aug 28 19:37:09 2003 (1062113829)
WARNING: solution did NOT reach the target value(s)

The mapped field inhomogeneity is 10.15 ppm
48.0x45.0 cm dsV : -4.96 to +5.20 (spec 12.00 PPM peak-to-peak) IN SPEC
40 cm dsV : -1.08 to +1.14 (spec 4.00 PPM peak-to-peak) IN SPEC
30 cm dsV : -0.28 to +0.32 (spec 1.25 PPM peak-to-peak) IN SPEC
20 cm dsV : -0.12 to +0.12 (spec 0.30 PPM peak-to-peak) IN SPEC

Unified Shim Info: G3UnifiedTarget
Target File : F:\FinalT\ShimTool\G3UnifiedTarget.to1

Unrounded: (Predicted)
48.0x45.0 cm dsV : 10.15PPM (-4.94/+5.21)
40 cm dsV : 2.22PPM (-1.06/+1.17)
30 cm dsV : 0.67PPM (-0.29/+0.38)
20 cm dsV : 0.31PPM (-0.15/+0.16)

Quantized: (Predicted)
48.0x45.0 cm dsV : 10.15PPM (-4.94/+5.21)
40 cm dsV : 2.22PPM (-1.06/+1.17)
30 cm dsV : 0.67PPM (-0.29/+0.38)
20 cm dsV : 0.31PPM (-0.15/+0.16)

```

SAMPLE SHIMTOOL TEXT FILE DISPLAYS
ILLUSTRATION 9-34

9-5 RUNNING SHIMTOOL (continued)

9-5-3 After ShimTool Processing (continued)

“.SHM” FILE
Showing Surface
Shimming Report

Shim Positions To Change
(Spec.: < 5)

Net Mass Change
(Spec.: < 5 mm)

Predicted PPM

Present (& Spec.) Values

48 x 45 EV Peak-to-Peak

45 cm DSV Peak-to-Peak

40 cm DSV Peak-to-Peak

30 cm DSV Peak-to-Peak

20 cm DSV Peak-to-Peak

Predicted Values
(After This Iteration)

48 x 45 EV Peak-to-Peak

45 cm DSV Peak-to-Peak

40 cm DSV Peak-to-Peak

30 cm DSV Peak-to-Peak

20 cm DSV Peak-to-Peak

```

SHIM OUTPUT FILE -- Generated By ShimTool

      SHIM MAGNET TYPE : Cylindrical
      MAGNET TYPE      : Corona
      AUTO-TOLERANCE   : 0.0000 @ 0.0000 PPM
      CRITICAL TARGET  : 0.0000 PPM
      INPUT FIELD FILE  : w0001600000AIS.map
      OUTPUT FILE       : C:\MRshim\w000160\w0001600000AIS.shm
      PREVIOUS FILE     : C:\MRshim\w000160\w0001600000AIS_5V.cur
      CALIBRATION FILE  : W_CoronaP.ca1
      ANALYSIS OPTION:  ADDITION AND SUBTRACTION (RE-SHIMMING)
      MINIMIZATION TECHNIQUE : ABSOLUTE

                                     ROUNDED
                                     -----
Total Number of Shim Pockets          268
Changed Number of Shim Pockets         2
Changed Number of SC Shims             4
Total Passive Shim Mass (shims)       1232.0000
Net Passive Shim Mass Change (shims)   -2.0000 (+0.0000/-2.0000)
Anticipated Field Tolerances (PPM)     10.592

AVERAGE FIELD (Gauss)                 Initial= 29993.98
                                       Final= 29993.96

FIELD TOLERANCE (PPM)
                                       Requested= Unified
                                       Final Before Rounding= 9.500

The mapped field inhomogeneity is ( 768 pts) 27.67
The reported field inhomogeneity is ( 768 pts) 27.67
NUMBER OF LP ITERATIONS                 = 701
RUN COMPLETED Thu Jan 22 17:30:42 2004 (1074821442)

The uncomp field inhomogeneity is 9.99 ppm
48.0x45.0 cm dsv : -4.84 to +5.15 (spec 26.00 PPM peak-to-peak) IN SPEC
45 cm dsv : -2.13 to +2.81 (spec 12.00 PPM peak-to-peak) IN SPEC
40 cm dsv : -1.08 to +1.30 (spec 4.00 PPM peak-to-peak) IN SPEC
30 cm dsv : -0.26 to +0.52 (spec 1.25 PPM peak-to-peak) IN SPEC
20 cm dsv : -0.00 to +0.33 (spec 0.30 PPM peak-to-peak) OUT OF SPEC

Unified Shim Info: G3Unified5V
Target File : 9.5,6.5,2,.3,.05

Unrounded: (Predicted)
48.0x45.0 cm dsv : 9.50PPM (-4.69/+4.81)
45 cm dsv : 4.56PPM (-2.02/+2.54)
40 cm dsv : 1.83PPM (-0.77/+1.06)
30 cm dsv : 0.30PPM (-0.09/+0.21)
20 cm dsv : 0.05PPM (+0.04/+0.09)

Quantized: (Predicted)
48.0x45.0 cm dsv : 10.59PPM (-5.25/+5.34)
45 cm dsv : 4.44PPM (-1.98/+2.47)
40 cm dsv : 1.90PPM (-0.83/+1.07)
30 cm dsv : 0.54PPM (-0.21/+0.33)
20 cm dsv : 0.17PPM (-0.02/+0.16)

```

SAMPLE SHIMTOOL TEXT FILE DISPLAYS
ILLUSTRATION 9-34 (CONTINUED)

9-5 RUNNING SHIMTOOL (continued)**9-5-3 After ShimTool Processing (continued)**

3. Record the “Anticipated Field Tolerance (PPM)” value recorded in the “.cur” file in DATA SHEETS, Table 6-2, under “Predicted PPM” for the current iteration.
4. Record the “Total Passive Shim Mass (shims)” value recorded in the “.cur” file in DATA SHEETS, Table 6-2, under “Passive Shim Mass (mm)” for the current iteration.

Note

Shim mass (shims) refers to the total thickness in millimeters of shims in all shim tray positions.

5. On the first iteration — using the ‘centered map’ and set to calculate S/C shimming only — look at the 4 volumes (“48.0 x 45.0 cm dsv,” “40 cm dsv,” “30cm dsv” and “20 cm dsv”) listed under “The recomp field inhomogeneity is” Compare these values to the specifications shown in Table 9-4.

Note

The “45 cm dsv” volume displayed when the ATR includes Surface Shimming is not important during field shimming and can be ignored.

- If **ALL** volumes **ARE** within specification, then the magnet is shimmed. Continue with Section 9-8, Wrap Up.
- If **ANY** volume under “The recomp field inhomogeneity is ...” **IS NOT** within specification, then compare the “Quantized (Predicted)” values for the 4 volumes to the specifications shown in Table 9-4.
 - If **ALL** volumes **ARE** within specification, then:
 - a. Apply the S/C shim currents called for by ShimTool in conformance with Section 9-7, Superconducting Shimming.
 - b. Take a map and run ShimTool again (Section 9-5) for S/C shimming ONLY.
 - c. Apply the S/C shim currents called for by ShimTool in conformance with Section 9-7, Superconducting Shimming.
 - d. Take a map and run ShimTool again (Section 9-5) for S/C shimming ONLY.
 - e. Continue with step 7.
 - If **ANY** volume **IS NOT** within specification:
 - a. Run ShimTool using the SAME map — starting with Section 9-5-2, Acquiring a Shim File — for S/C shimming ONLY.
 - b. Apply the S/C shim currents called for by ShimTool in conformance with Section 9-7, Superconducting Shimming.
 - c. Take a map and run ShimTool again (Section 9-5) for **BOTH** passive and S/C shimming.
 - d. Continue with step 6.

9-5 RUNNING SHIMTOOL (continued)**9-5-3 After ShimTool Processing (continued)**

TABLE 9-4
VOLUMETRIC SPECIFICATIONS

Volume	Peak-to-Peak PPM Specification
48 x 45 cm (elliptical) DSV	12.00 ppm
40 cm DSV	4.00 ppm
30 cm DSV	1.20 ppm
20 cm DSV	0.30 ppm

6. After running ShimTool set for both passive and S/C shimming, look at the “Net Mass Change” and the “Changed Number of Shim Positions” values ShimTool reports (Illustrations 9-31, 9-33 and 9-34). Compare these values to the specifications shown in Table 9-5.

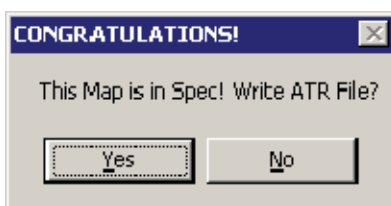
TABLE 9-5
PASSIVE SHIM CHANGE SPECIFICATIONS

Measure	Specification
Net Mass Change	< 5 mm
Changed Number of Shim Positions	< 5 Positions

- **If the specifications in Table 9-5 HAVE NOT been met** (either “Net Mass Change” is ≥ 5 mm **OR** “Changed Number of Shim Positions” is ≥ 5), then:
 - a. **FIRST** install the passive shims called for by ShimTool in conformance with Section 9-6, Passive Shimming.
 - b. Once the passive shims have been installed, apply the S/C shim currents called for by ShimTool in conformance with Section 9-7, Superconducting Shimming.
 - c. Take a map and run ShimTool again (Section 9-5) for **BOTH** passive and S/C shimming.
 - d. Repeat step 6.
- **If the the specifications in Table 9-5 HAVE been met** (both “Net Mass Change” is < 5 mm **AND** “Changed Number of Shim Positions” is < 5), then:
 - a. Take a map and run ShimTool again (Section 9-5) for S/C shimming **ONLY**.
 - b. Apply the S/C shim currents called for by ShimTool in conformance with Section 9-7, Superconducting Shimming.
 - c. Take a map and run ShimTool again (Section 9-5) for S/C shimming **ONLY**.

9-5 RUNNING SHIMTOOL (continued)**9-5-3 After ShimTool Processing (continued)**

7. Look at the text “.cur” file’s 4 volumes (“48.0 x 45.0 cm dsv,” “40 cm dsv,” “30 cm dsv” and “20 cm dsv”) listed under “The recomp field inhomogeneity....” Compare these values to the specifications shown in Table 9-4.
 - Run ShimTool again (Section 9-5) for S/C shimming (ONLY) **if ANY volume IS NOT within specification.**
 - If ALL volumes **ARE** within specification, then the screen shown in Illustration 9-35 will appear.
 - a. Select **[No]**.
 - b. Exit ShimTool and continue with Section 9-8, Wrap Up.



THE “CONGRATULATIONS” SCREEN
ILLUSTRATION 9-35

Note

The “Congratulations” screen means the magnet has been shimmed to within specification. Writing an ATR file is not needed after the magnet has left the factory.

9-6 PASSIVE SHIMMING

WARNING!

STEEL SHIMS IN A MAGNETIC FIELD ARE SUBJECT TO A FORCE PROPORTIONAL TO THE STEEL'S MASS AND TO THE FIELD'S STRENGTH. IN A 3 TESLA FIELD FORCES > 50 POUNDS (22.6 KG) MAY BE ENCOUNTERED:

- AN ATTRACTION FORCE PULLING A SHIM TRAY INTO THE MAGNET WILL BE ENCOUNTERED WHEN A LOADED TRAY IS BROUGHT WITHIN 3 FEET (1 METER) OF THE MAGNET BORE.
- AN ATTRACTING AND / OR REPELLING FORCE WILL BE ENCOUNTERED WHEN INSERTING / REMOVING A LOADED SHIM TRAY INTO / FROM A TRACK INSIDE THE MAGNET BORE.
- A REPELLING FORCE ON A SHIM TRAY INSIDE THE MAGNET'S BORE MAY PROPEL THE TRAY OUT FROM THE MAGNET WHEN THE TRAY IS RELEASED FROM ITS FIXED POSITION.

MAKE SURE TO FOLLOW THESE SAFETY REQUIREMENTS DURING PASSIVE SHIM INSTALLATION / REMOVAL:

- ASSEMBLE ALL EQUIPMENT AND CHECK FUNCTIONALITY IN CONFORMANCE WITH THE PASSIVE SHIM INSTALLATION PROCEDURE.
- LOAD SHIM TRAYS OUTSIDE THE MAGNET ROOM. NEVER HAVE LOOSE INDIVIDUAL SHIMS INSIDE THE MAGNET ROOM.
- SECURE SHIMS IN SHIM TRAYS AND INSTALL TRAYS INTO THE MAGNET IN STRICT CONFORMANCE WITH THE PASSIVE SHIM INSTALLATION PROCEDURE.
- WEAR SAFETY GOGGLES / GLASSES DURING SHIM INSTALLATION / REMOVAL TO PREVENT EYE INJURY FROM SUDDEN TRAY MOVEMENT.
- WEAR LEATHER GLOVES DURING SHIM TRAY INSTALLATION / REMOVAL TO PREVENT SKIN INJURY FROM FIBERGLASS SLIVERS.
- HOLD SHIM TRAYS FIRMLY IN PLACE WHEN RELEASING INSTALLED TRAYS. MAKE SURE ALL LOADED SHIM TRAYS ARE RESTRAINED (BY HAND, SHIM TRAY PULLER OR FIXED INTO ITS TRACK), ESPECIALLY TRAYS MARKED ON THE END FLANGE AS MOUNTED IN AN OUTWARD REPELLING FORCE.
- INSERT / REMOVE SHIM TRAYS WITH A GRADUAL MOTION.
- TWO PERSONS ARE REQUIRED TO INSTALL / REMOVE TRAYS.
- DO NOT LOAD SHIM TRAYS WHILE ANYONE IS DIRECTLY IN LINE WITH THE TRAY ON EITHER END OF THE MAGNET.
- NEVER POSITION YOURSELF BETWEEN A SHIM TRAY AND THE MAGNET.

9-6 PASSIVE SHIMMING (continued)

IMPORTANT !!!

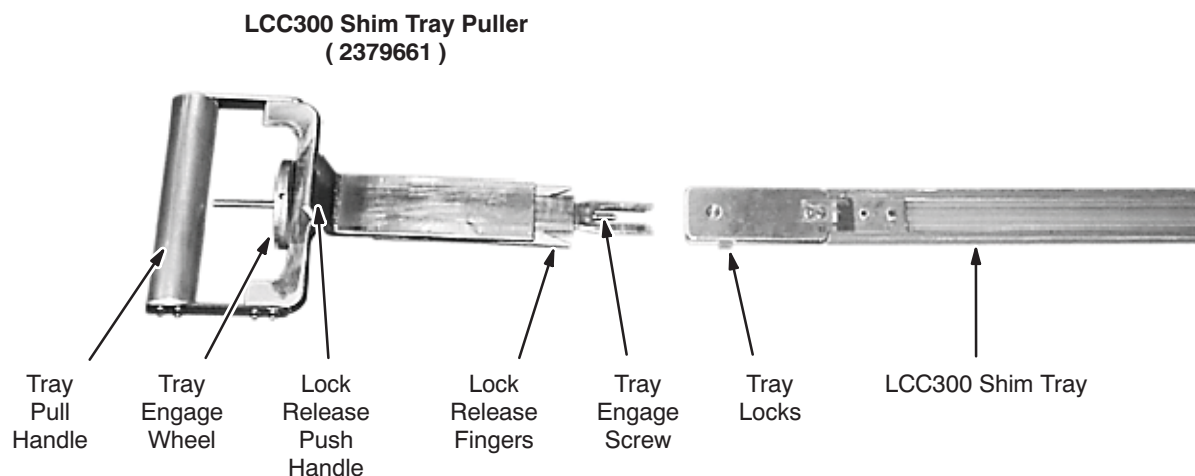
The LCC300 Field Mapping Fixture's patient end support bracket will have to be detached from the magnet in conformance with SET-UP AND CALIBRATION, Section 5, Preparation for Field Measurement, while accessing the shim trays behind it and reattached before further mapping.

9-6-1 Shim Tray Removal

CAUTION

Wear non-slip gloves during shim tray insertion / removal to protect against fiber-glass splinters from the trays.

1. Align the LCC300 Shim Tray Puller (2379661) with the shim tray to be pulled. See Illustrations 9-36.

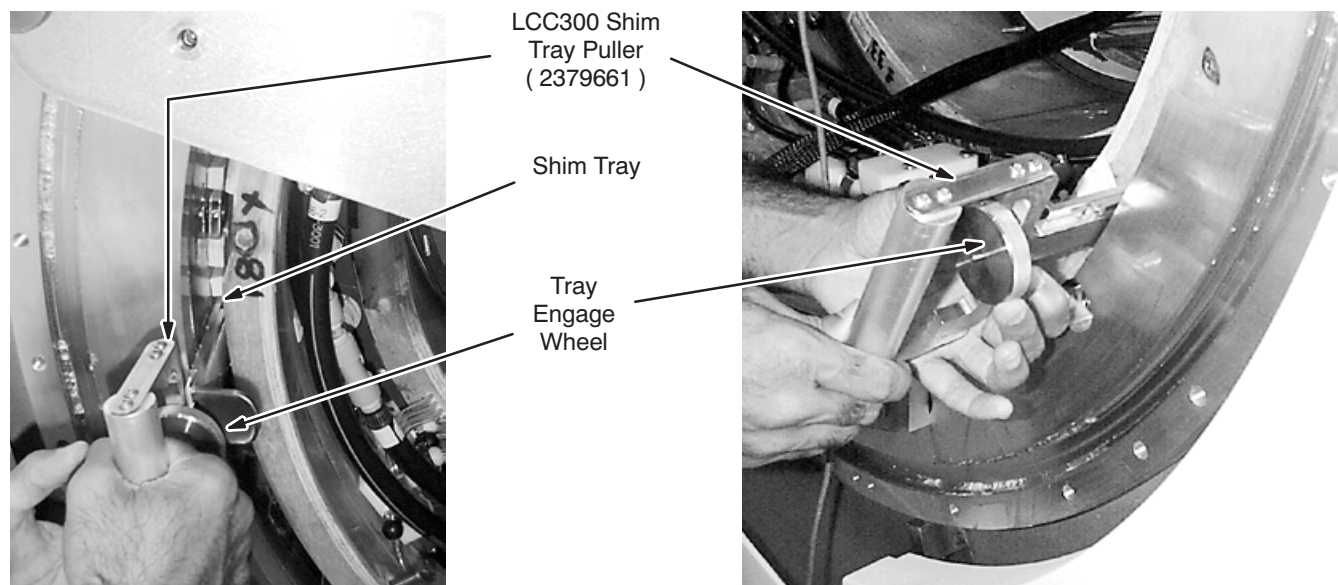


ALIGN SHIM TRAY PULLER WITH SHIM TRAY
ILLUSTRATION 9-36

2. Insert the shim tray puller's tray engage screw into the threads at the end of the shim tray, then turn the puller's tray engage wheel to fully engage the screw with the tray. See Illustration 9-37.

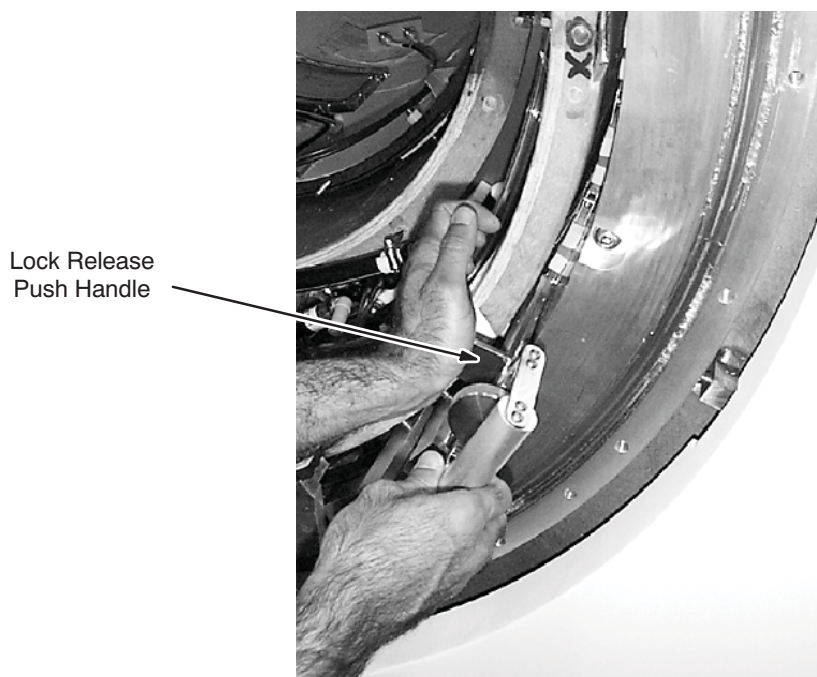
IMPORTANT !!!

The shim tray puller **MUST** be fully and solidly engaged with the shim tray.

9-6 PASSIVE SHIMMING (continued)**9-6-1 Shim Tray Removal (continued)**

ENGAGE SHIM TRAY PULLER WITH SHIM TRAY
ILLUSTRATION 9-37

3. Push the shim tray puller's push handle to release the tray's locking prongs. See Illustration 9-36 and 9-38.



RELEASE SHIM TRAY LOCKS
ILLUSTRATION 9-38

9-6 PASSIVE SHIMMING (continued)**9-6-1 Shim Tray Removal (continued)****CAUTION**

Be prepared to manage the effects of a 3.0T field on the shim tray during tray removal / insertion. Adhere to all safety material stated on page 9-50.

CAUTION

Wear gloves when handling Shim Trays to protect hands from fiberglass slivers.

- Carefully slide the shim tray straight out from its track. Have the work assistant hold onto the tray as soon as possible.



SHIM TRAY REMOVAL
ILLUSTRATION 9-39

- Work Assistant: Securely hold the shim tray and carry it and the puller away from the magnet's field to a shim tray loading area in another room.

9-6 PASSIVE SHIMMING (continued)

9-6-2 Loading Passive Shims

IMPORTANT !!!

Load shims into the shim tray in a room away from the magnet's field.

1. **First person:** Identify which component shims are required to meet the total shim mass specified by ShimTool for each position in each tray. See Illustration 9-33 for a sample ShimTool shimming report, Illustration 9-40 for the LCC300 shim tray layout and Illustration 9-41 for assembly of component shim per shim position.

IMPORTANT !!!

Make sure screws holding shims to shim tray are fully tightened. Use Torque Screwdriver 46-294167P81. Loose screws / shims can cause white pixels.

Note

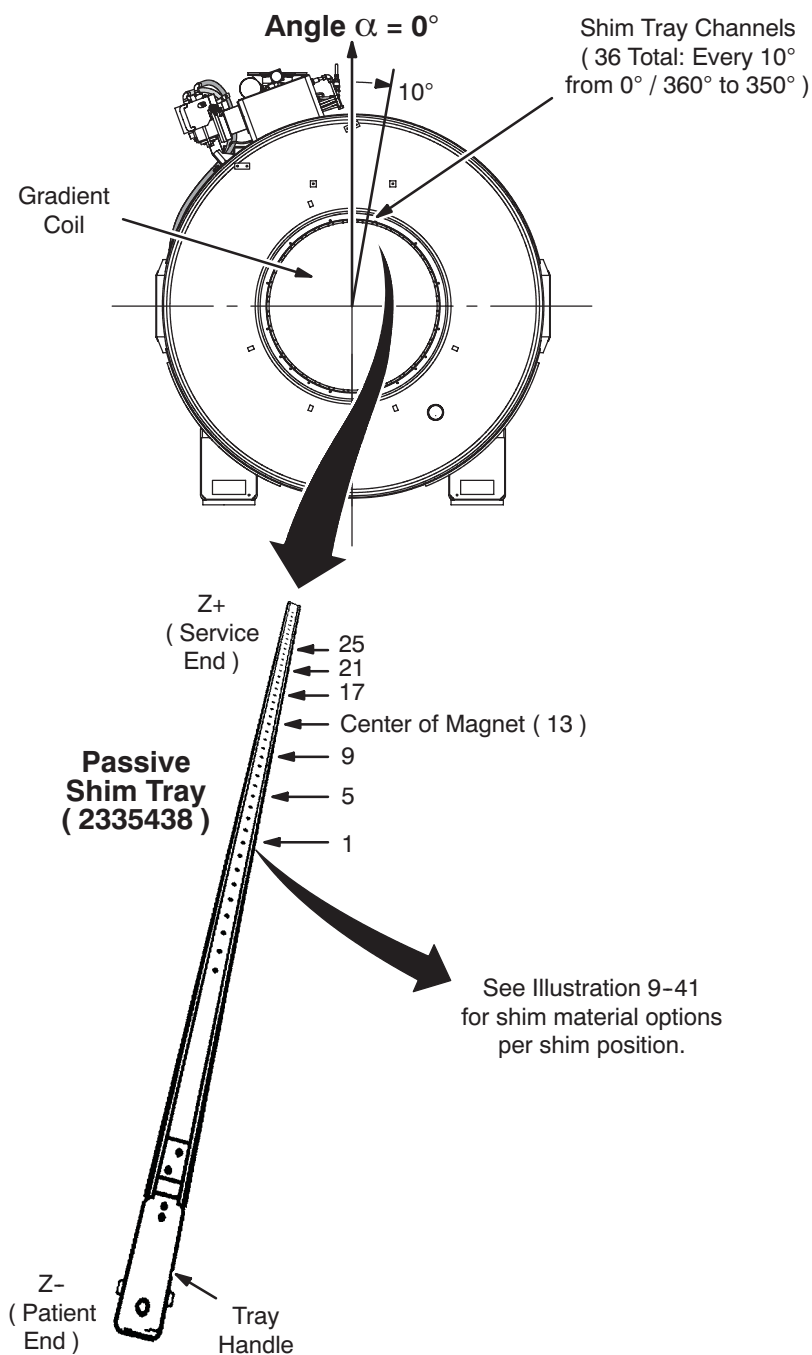
Make sure half-width 2 mm shims are oriented across the shim tray as shown in Illustration 9-41.

2. Securely screw the component shim arrangement to the shim tray position as shown in Illustration 9-40. Torque each screw to 10 inch-pounds using the torque screwdriver (46-294167P81) included in LCC300 Passive Shimming Kit 2386029. Orient every position in each tray identically (screw heads facing the same way, etc.).
3. **Second person (NOT the first person):** Verify that incremental shims have been added / subtracted in conformance with the "Passive Shims - Incremental Rounded Shim" information on page 3 of ShimTool's printout (Illustration 9-33).
4. Both people separately verify that the total shims at each tray position equals the total shims for that tray position specified in the "Passive Shims - Total Rounded Shim" information on page 4 of ShimTool's printout (Illustration 9-33).
 - If the actual shim totals in each position match ShimTool's printout / ".shm" file, then continue with Section 9-6-3, Shim Tray Insertion.
 - If the actual shim totals do not match ShimTool's printout / ".shm" file in any position, then:
 - a. Edit this iteration's ".shm" file's "Total - Quantized" pages to reflect the actual steel on the trays.
 - b. Save the edited ".shm" file using its **existing** name.

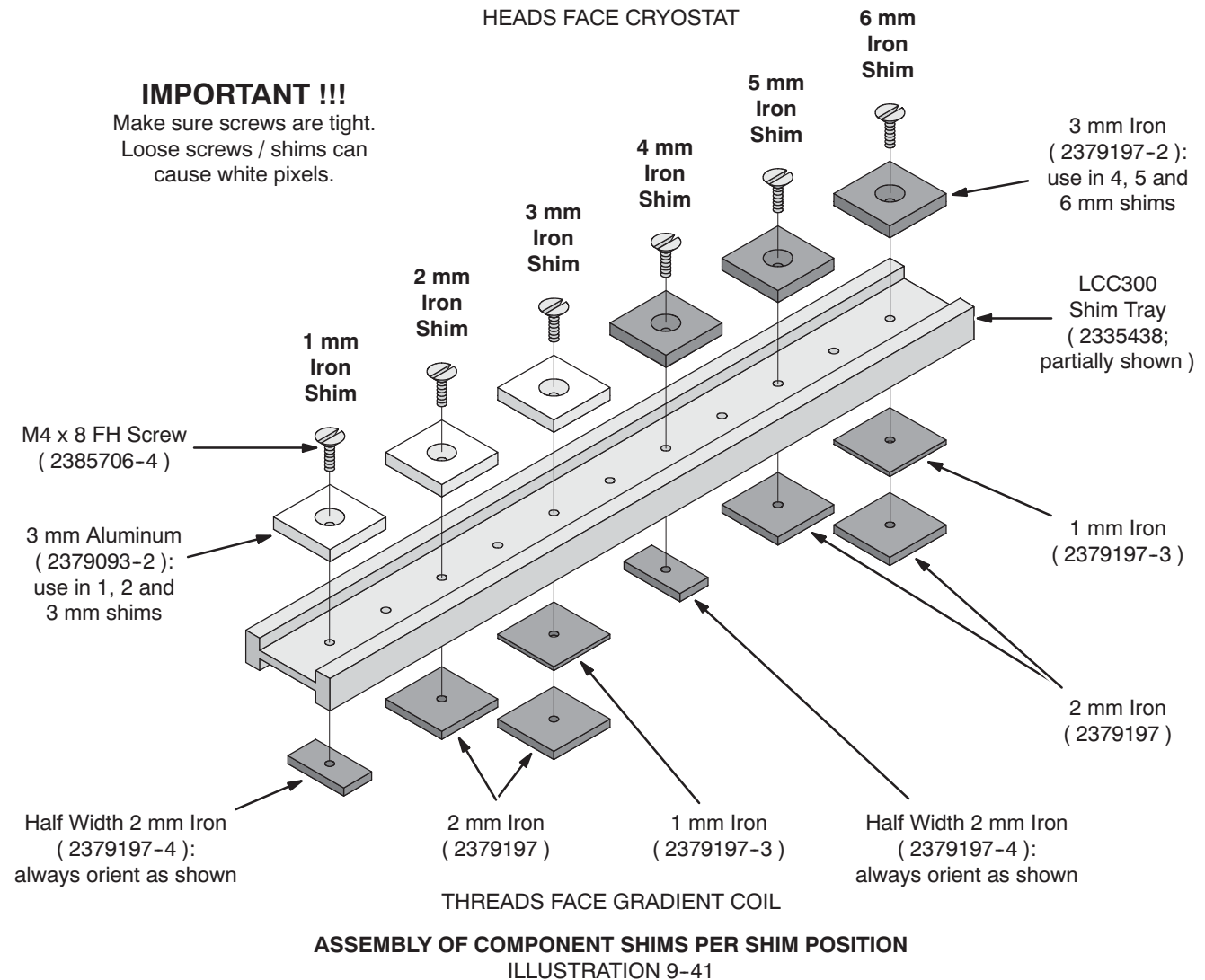
IMPORTANT !!!

Do NOT change the name of the ".shm" file. During the next shimming iteration ShimTool uses the steel quantity / location information from this iteration's ".shm" file and a new map of the field with that steel installed to calculate its new report.

- c. Continue with Section 9-6-3, Shim Tray Insertion.

9-6 PASSIVE SHIMMING (continued)**9-6-2 Loading Passive Shims (continued)**

SHIM TRAY LAYOUT
ILLUSTRATION 9-40

9-6 PASSIVE SHIMMING (continued)**9-6-2 Loading Passive Shims (continued)****9-6-3 Shim Tray Insertion**

1. Screw the shim tray puller to the end of the shim tray.



Be prepared to manage the effects of a 3.0T field on the shim tray during tray removal / insertion. Adhere to all safety material stated on page 9-50.

2. Securely carry the puller and tray to the magnet.

9-6 PASSIVE SHIMMING (continued)**9-6-3 Shim Tray Insertion (continued)**

3. Align the end of the shim tray with the tray's track. Orient every tray in conformance with Illustration 9-42.

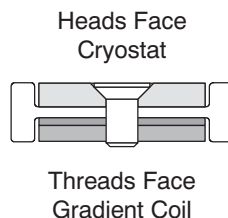
**PROPER SHIM TRAY ORIENTATION**

ILLUSTRATION 9-42

4. Carefully slide the shim tray straight into its track. Have the work assistant assist until enough hand-hold is no longer available.
5. Pull the shim tray puller's push handle back to allow the tray's locking prongs to reengage.
6. Turn the shim tray puller's wheel until the puller is disengaged from the tray.

9-7 SUPERCONDUCTING SHIMMING

WARNING!

MAKE SURE THE MAGNET ROOM EXHAUST VENT FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.

IMPORTANT !!!

Make sure that the shim power supply has been calibrated in conformance with the vendor's manual and that the shim heater currents are set as shown below before use.

Transverse 1 Shim Heater Current	610 ± 5 mA
Transverse 2 Shim Heater Current	610 ± 5 mA
Axial Shim Heater Current	710 ± 10 mA

Note

Always check ATR for any possible magnet-specific current settings.

9-7-1 Introduction

The "Recommended Total Current" and polarity values calculated by the shim program must be entered into the S/C Correction Coils before performing subsequent field mapping.

The coil groups are:

- Transverse Coils, Odd: T1-1 through T1-5
- Transverse Coils, Even: T2-1 through T2-5
- Axial Coils: AX1 through AX4

Each coil group is connected to one heater circuit on the shim power supply. Therefore, if the only current change was in the Axial 2 Coil, turning the Axial Switch Heater on would dump the currents in Axial Coils (1 - 4) and the appropriate currents for each Axial Coil would need entering.

9-7 SUPERCONDUCTING SHIMMING (continued)

9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming

WARNING!

BEFORE TURNING ON THE SHIM COIL POWER SUPPLY AND APPLYING SHIM CURRENTS, MAKE SURE THE FOLLOWING CONDITIONS EXIST TO PREVENT IRREPARABLE DAMAGE TO THE VAPOR-COOLED SHIM LEADS:

- THE ORIFICE ON THE SHIM LEAD VENT IS IN PLACE.
- THE SHIM LEAD EXTENSION IS FROSTED AND VENTING.
- THE POWER SUPPLY CURRENT CONTROLS ARE SET TO ZERO (FULL CCW).

CAUTION

Whenever the Shim Lead is to be engaged, the NuPro Valve needs to be opened before and during engagement to avoid a possible quench. Wait for the valve to frost, then slowly engage the Shim Lead. Do not remove the orifice cap on the Shim Lead that controls the gaseous helium flow. See SET-UP AND CALIBRATION, Section 6, Shim Lead Engage / Disengage.

1. Make sure vapor is escaping from the Shim Lead and the lead is frosted then turn on the power supply.
2. Repeat the following steps (a - h) to input shim currents for each shim group in sequence (T1, then T2 and finally Axial). See Table 9-6.
 - a. Set the Shim Group Select Switch to the appropriate group (T1, T2 or Axial). See Illustration 9-43.

9-7 SUPERCONDUCTING SHIMMING (continued)**9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)****2. (continued)****Note**

The present current / polarity, referred to in the following steps, is the current / polarity that presently exists in each shim coil. The present currents / polarities are those ShimTool calculated / reported (in the “Previous” column under “Active Shim Currents” in Illustration 9-33) in its previous iteration, and then were loaded into the shim coils and recorded in DATA SHEETS, Table 6-1, in the “New” column.

IMPORTANT !!!

On a reverse polarity magnet enter negative (–) polarity currents where ShimTool says positive (+) and positive (+) polarity currents where the ShimTool says negative (–).

- b. Match the Shim Power Supply current / polarity of each channel (coil) in each group to the present current / polarity in the selected shim coil group. See Illustration 9-43.
- Set the Polarity Toggle Switches (up = normal / positive polarity; down = reversed / negative polarity) as required for to match the present polarity for the each coil in the group.
 - Set the Course / Fine Current Adjust potentiometers on the power supply to match the present current level for the each coil in the group.

Note

The polarity relays will not change polarity if there is any current flowing through their respective channels. The current must be reduced to zero before the relays will change to the position indicated by the external switches on the power supply.

TABLE 9-6
S/C SHIM GROUPS

Course / Fine Adjust Potentiometer & Polarity Light / Switch	'Shim Group Select' Switch Setting		
	Transverse 1 Shim Group (Middle Switch Position)	Transverse 2 Shim Group (Bottom Switch Position)	Axial Shim Group (Top Switch Position)
1	T1-1 (ZX^2Y^2)	T2-1 (ZXY)	AX1
2	T1-2 (X)	T2-2 (Y)	AX2
3	T1-3 (Z^2X)	T2-3 (Z^2Y)	AX3
4	T1-4 (ZX)	T2-4 (ZY)	AX4
5	T1-5 (X^2Y^2)	T2-5 (XY)	—
6	—	—	—

9-7 SUPERCONDUCTING SHIMMING (continued)**9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)**

2. (continued)

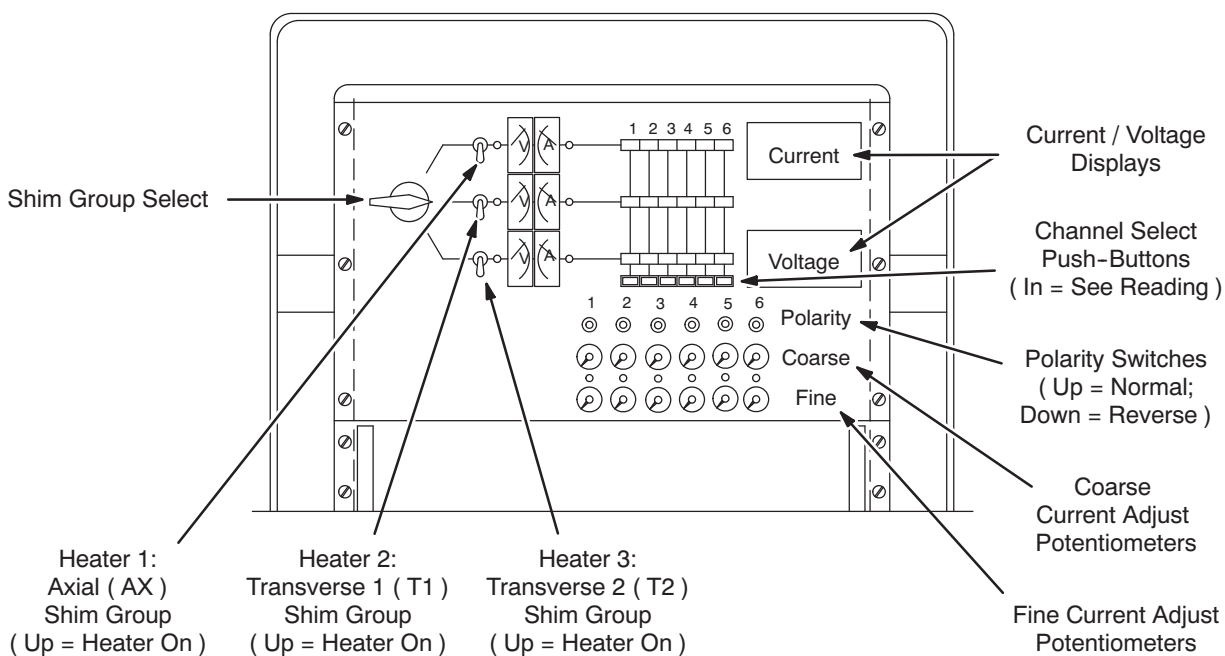
**SHIM POWER SUPPLY CONTROLS (FRONT PANEL)**

ILLUSTRATION 9-43

9-7 SUPERCONDUCTING SHIMMING (continued)**9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)**

2. (continued)

CAUTION

When the Switch Heaters are turned on, any existing currents in the Shim Coils will be discharged into the power supply. To prevent dumping excessive currents through the Shim Leads, match the existing shim currents at the power supply with those in the magnet before turning on the Switch Heaters. The current then can be adjusted to the required new levels after the Switch Heaters are activated.

CAUTION

Magnet damage is possible if heater currents are not set correctly.

- c. Verify that all shim power supply currents / polarities match the currents / polarities in the selected shim coil group.
- d. Turn on the selected group's Switch Heater (Transverse 1, Transverse 2 or Axial) and wait 30 seconds for the switches to go resistive.

Note

If there is a polarity change from the present to the new current, set the current to zero before changing the polarity switch.

- e. Set the new "total" currents and polarities for the shim coil group using ShimTool's latest "Active Shim Currents" values (i.e., the currents listed in the column labelled "New" on the "Active Shim Currents" page of the ShimTool report shown in Illustration 9-33).

9-7 SUPERCONDUCTING SHIMMING (continued)**9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)**

2. (continued)

IMPORTANT !!!

The LCC300 magnet correction coils have greater inductances than those of a 1.5T magnet. Therefore, a longer settling time is required for the shim currents. It is important that the shim current settling time is in conformance with Table 9-7 to prevent subsequent current drift after settling.

- f. Allow the shim currents to settle for the time shown in Table 9-7.

TABLE 9-7
SHIM CURRENT SETTLING TIMES

Shim Coil Group	Settling Times
T1	10 minutes
T2	
Axial*	20 minutes
* Axial times represent minimum times. Axial coupling may require longer wait times to meet the field stabilization requirements stated in Step 4 below.	

- g. Verify the new currents, then turn off the heater group and wait 5 minutes.
- h. After waiting 5 minutes, gradually turn the power supply Current Adjust Potentiometers down to zero.

Note

If repeated iterations of the shim program call for the same current increment in a shim coil, indicating the Shim Switch not going resistive, then one of the two following actions may help drive the Shim Switch resistive:

- Make sure the Heater current is set properly **OR**
- Set 20 amps in the coil, then lower the current to the required value.

Adjust slightly, as required. 1 mA high is better than 1 mA low.

Note

Failure to accurately set the power supply currents and polarity can greatly increase the number of plotting iterations (i.e., time to shim the magnet).

9-7 SUPERCONDUCTING SHIMMING (continued)

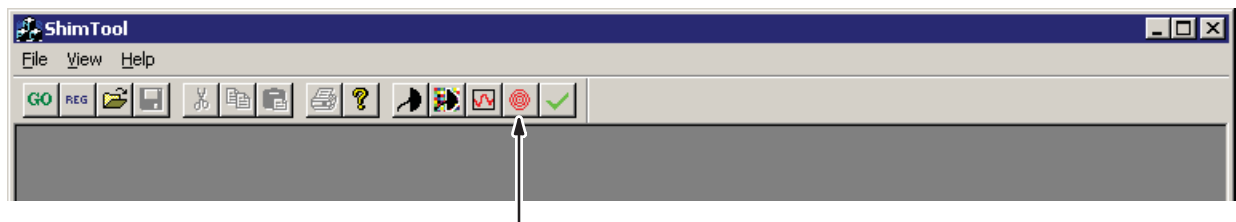
9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)

3. Make sure the field camera is in the 0° position.
4. For axial currents (**ONLY**), monitor field drift until it stabilizes (i.e., < 2 Hz change in total field over a two minute period):

Note

In some cases, the frequency could vary as much as 100 Hz. Shutting off the Coldhead Compressor during acquisition / mapping of shim data may help reduce the frequency change.

- a. Choose the **[Continuously Monitor Field]** button (second button from right) from ShimTool's main screen. See Illustration 9-44.



Continuously Monitor Field
Button

SHIMTOOL MAIN SCREEN BUTTONS

ILLUSTRATION 9-44

- b. Select "Probe 22."
- c. Select **[Start]** on the screen that appears to begin measuring field strength and drift. Make note of the starting frequency for use in step 4d.
- d. Wait 2 minutes, then select **[Stop]** and record the change between the starting and present frequencies.
 - If the change is < 2 Hz, continue with step 5.
 - If the change is $\geq$ 2 Hz, repeat steps 4c and 4d.
5. Once the field is stable, turn off the last Switch Heater that was turned on and allow the Heater to cool for 5 minutes.
6. Slowly turn all the Course / Fine Current Adjust potentiometers on the shim power supply back down to zero (0) amps (full CCW).

9-7 SUPERCONDUCTING SHIMMING (continued)**9-7-2 Adjusting S/C Shim Coil Power Supply Currents for Active Shimming (continued)**

Make sure the Shim Switch is in the Transfer 1 or Transfer 2 position before turning off the Shim Power Supply to prevent any random quench of the axial shim currents when the Shim Power Supply is turned off.

7. When s/c shimming is complete, turn off the power supply, then disconnect all leads between the magnet and the Shim Power Supply (P2, P2W, Heater Cable).
8. Close the Shim Lead Nupro Vent Valve.



Make sure the cap is replaced and does not leak and result in gaseous helium loss / frosting.

9. Retract the Shim Lead, tighten the brass cap and check for leaks.

9-8 WRAP UP

1. Archive the “.map”, “.shm” and “.cur” files by sending them to the On-Line Center in conformance with On-Line Center instructions.
2. Perform SET-UP AND CALIBRATION, Section 5-6, After Field Mapping, to put the magnet back in normal condition.

Note

For Field Adjustments after Shimming see SET-UP AND CALIBRATION, Section 10. For Magnetic Field Stability Checks see FUNCTIONAL CHECKS, Section 4.

